Energy Shocks and the Green Adoption Channel in a Small Open Economy

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Abstract

European governments spent more than €600 billion shielding households from the 2021–23 energy shock, mostly by suppressing retail energy prices (Bruegel, 2023). We show that price suppression carries a cost the macroeconomic literature has missed: it switches off the crisis-time adoption of green durables. We add a discrete brown/green durable choice to the small open economy HANK model of Auclert et al. (2024). An energy shock raises the relative price of brown energy and triggers an adoption wave: a peak green-share response of about nine percentage points from a five-percent base. Because the switch is a lumpy purchase financed against a borrowing constraint, adoption concentrates in constrained households and runs through the relative price alone. A retail price cap, which freezes that price, eliminates the wave; an income transfer of equal fiscal cost, which leaves the price free, preserves it, at comparable output stabilisation. The same relative-price suppression also shuts down the expenditure-switching gain a small open economy draws from the shock. Two further results follow: an untargeted flat transfer dominates one indexed to pre-crisis energy use, because homothetic energy demand makes energy-indexed targeting regressive; and a permanent carbon tax that greens the steady state provides no first-order crisis insurance. Monetary policy moves the transition only weakly, through the financing cost of the switch. If governments shield, they should shield incomes, not prices.

Keywords: energy shocks; green transition; durable adoption; HANK; price caps; transfers; carbon taxation.

JEL codes: E21, E32, F41, H23, Q43, Q58.

1 Introduction

When wholesale energy prices tripled in 2021–22, European governments reached for shields. Between September 2021 and January 2023 EU member states committed more than €600 billion to protect households and firms, most of it through measures that hold down the retail price itself: caps, regulated tariffs, and energy tax cuts (Bruegel, 2023). The shields worked on their own terms; they protected purchasing power. But they act on the same object the transition depends on. A retail energy price that is held fixed cannot signal scarcity, and the price signal is the mechanism through which households substitute away from fossil energy. The institution that assembled the spending figures made the point directly, urging governments to replace price-suppressing measures, which it called de facto fossil-fuel subsidies, with targeted income support (Bruegel, 2023).

This paper isolates one margin of that trade-off the macroeconomic literature has left out: the crisis-time adoption of green durables. A fossil price spike is exactly when switching to a technology that does not burn fossil energy pays best, and European households moved. Heat-pump sales reached a record of about three million units in 2022, up roughly 38% as gas prices surged; they fell 5% in 2023 and a further 21% in 2024 as gas normalised, the industry association naming the low price of subsidised gas among the three main causes, and recovered only once governments cut electricity taxes (EHPA, 2023, 2024, 2025). The margin is real, it tracks the relative price, and the shields reached it.

We put that margin into an otherwise standard heterogeneous-agent model of an energy-importing economy. Households in the small open economy HANK of Auclert, Monnery, Rognlie, and Straub (2024; henceforth AMRS) choose discretely whether to pay a one-time resource cost for a green durable that insulates them from the fossil price, a heat pump or an electric vehicle, modelled as a logit adoption choice over the durable stock and solved in the sequence space. Everything else is inherited from Auclert et al. (2024): the CES energy nest, sticky import prices, the wage Phillips curve, the constant-real-rate monetary baseline. The aggregate block is a deliberately generic energy importer; the only new object is the extensive margin, and the European episode is the economy we speak to (Section 4, Section 8). The closest antecedent, Dietrich et al. (2025), shows in a representative-agent model that a carbon-pricing transition is mildly inflationary and green durables are interest-rate sensitive; we keep that channel but add heterogeneity, an energy shock, and the fiscal and open-economy blocks, and ask not how a transition unfolds but how a crisis and the policies that meet it move the adoption margin.

The central result is a non-neutrality. An energy shock endogenously triggers an adoption wave: in the baseline crisis the green share rises by about nine percentage points at the peak, from a five-percent base. The two ex-post instruments in the recent literature, a retail price cap and a Slutsky-compensating transfer indexed to pre-crisis energy use, stabilise output comparably, but they part on the new margin. The cap holds the brown retail price at its pre-crisis level and thereby switches adoption off: the peak green-share response falls from nine points to essentially zero. The transfer supports incomes and leaves the wave intact. Shielding consumers from the relative price is not neutral for the transition, and the cost is not a marginal distortion of intensive-margin demand: it is the extensive margin itself. The mechanism runs through the brown retail price alone, which the cap fixes and the transfer leaves free, so the ranking survives every other choice in the model (Section 6).

Who should be shielded, and how, is the second question. Because energy demand is close to proportional to total consumption within technology groups, indexing a transfer to pre-crisis energy use hands the largest cheques to the wealthiest, lowest-MPC households, where a euro buys the least insurance. An untargeted flat transfer of the same fiscal cost reaches the high-MPC households the crisis hits hardest and delivers strictly higher welfare (a consumption-equivalent variation of -0.32% against -0.82%), while leaving adoption intact. The result is conditional on homothetic energy demand: with a declining Engel curve for energy, the mechanism of Bayer et al. (2026) through energy-share heterogeneity and of Langot et al. (2026) through a subsistence floor, low-income households carry a larger energy share and the ranking could weaken. We take the homothetic nest from Auclert et al. (2024) and leave that case aside. Under constant energy shares, energy-indexed targeting is dominated by an equal-cost flat transfer.

The third question turns from ex-post to ex-ante policy. Europe is about to run the ex-ante version of this experiment on purpose: the forthcoming ETS2 carbon price for buildings and road transport will raise the household brown-energy price deliberately (European Commission, 2026). A permanent carbon tax that greens the steady state before the crisis is a poor substitute for ex-post insurance: its crisis dividend is zero to first order, and it does not improve as shocks grow persistent. The prepared economy in fact adopts *more* during the crisis; what it lacks is not switching volume but the welfare value of the marginal switch, because pre-greening has already spent an adoption absorber that pays off precisely when the economy enters the crisis with room to green (Section 7). The case for the carbon tax is the externality and the standing gain of a greener steady state, not insurance against the next shock.

Two auxiliary results are genuinely conditional, and we flag them rather than resolve them. The sign of the cap's private-welfare effect turns on the energy-supply elasticity, which locates one axis of the disagreement between Langot et al. (2026) and Bayer et al. (2026) without nesting their models. The sign of the adoption channel's contribution to output turns on how the green resource flows are booked in the balance of payments; we report the import booking throughout and collect the alternative in Appendix A. The policy ordering is robust to both, since it follows from the relative price. The size of the adoption wave, by contrast, rests on one parameter, the logit taste scale σ_ε, and we discipline it rather than assume it. At a five-percent green share every brown household sits in the exponential tail of the logit, so the semi-elasticity of switching is $1/\sigma_\varepsilon$ household by household (Appendix D). Matching the response of adoption to a \$1,000 incentive in the quasi-experimental vehicle literature (Gallagher and Muehlegger, 2011; Chandra et al., 2010; Muehlegger and Rapson, 2022) pins $\sigma_\varepsilon \in [0.05, 0.07]$, with an implied green premium of \$11,000–\$14,000. The steady state and the ordering are invariant to the scale; the size of the wave is what it sets.

Section 2 places the paper in the literature. Section 3 presents the model and Section 4 its calibration, including the identification of the taste scale. Section 5 dissects the adoption channel under price and supply shocks, Section 6 compares the cap and the transfers, Section 7 studies ex-ante carbon pricing, Section 8 maps the results into European policy, and Section 9 concludes.

2 Related literature

We draw on three bodies of work: energy-shock stabilisation in heterogeneous-agent open economies; the macroeconomics of climate policy and household technology adoption; and the microeconometrics of durable adoption that disciplines the margin we add.

2.1 Energy-shock stabilisation in heterogeneous-agent economies

The closest work studies fiscal and monetary responses to the 2021–23 energy shock in open HANK economies. Auclert et al. (2024) provide the aggregate frame we inherit, an energy-importing HANK with a CES energy nest, sticky import prices, and intertemporal energy suppliers, and show that a low substitution elasticity makes the shock contractionary through the real-income channel, so a subsidy stabilises output, inflation and inequality at once; its drawback in their world is a cross-country externality, since coordinated subsidies spike the world price. Bayer et al. (2026) build a two-country monetary union to study exactly that spillover, with a richer medium-scale block, two assets, capital, energy in production, and a distributional mechanism that runs through within-country energy-share heterogeneity: poorer households spend more on energy and suffer larger real-income losses. Langot et al. (2026) take France as a small economy inside the euro area, fed by the government's official energy-price forecasts, with a higher intensive substitution elasticity and a subsistence energy floor that makes demand non-homothetic; on a conditional-forecast path they find the French shield effective, largely through a labour-cost channel that contains the price–wage spiral.

These models disagree on the sign of the shield's welfare effect, and the disagreement has an axis. Under perfectly elastic supply (Langot et al., 2026) the shield is effective; under a fixed union-wide energy quantity (Bayer et al., 2026) price suppression is destructive, because it blocks the substitution scarcity requires. Our two supply closures, elastic price-taking and fixed-quantity, reproduce the sign each obtains, which locates one source of the contrast in a single parameter, the supply elasticity at the relevant horizon. We do not over-read this. With no second country we cannot represent Bayer's cross-border transmission, and our wage block, lacking an investment or employment margin, reproduces the sign of Langot's pro-shield result but not its full labour-cost mechanism.

What none of these models contains is an endogenous energy technology. AMRS, Bayer, and Langot hold the household's energy type fixed, so the shield's cost to the transition is invisible to them. The nearest antecedent on the technology margin is Audzei and Sutóris (2026), who add continuous energy-conservation capital to a stylised heterogeneous-agent New Keynesian economy and show that the monetary stance shapes households' incentive to build resilience against energy price shocks. Their margin is an *intensity* of energy use: there is no fossil/green technology type, hence no relative energy price for a cap to suppress and no composition of the durable stock for fiscal policy to move. Our contribution is to make that type a choice. The adoption margin adds a cost of price suppression the aggregate literature cannot see, capping the brown price switches the transition off, without displacing the AMRS real-income channel, which operates in the background of every experiment. Our heterogeneity is over discount factors and the adoption margin rather than over energy shares, so we are silent on the energy-share-of-the-poor mechanism; the two distributional channels are complements, not substitutes.

2.2 Climate policy and household technology adoption

The second literature studies the macroeconomics of climate policy, and in particular its interaction with monetary policy. Governments have pursued opposite strategies to move households onto clean durables: the United States leans on subsidies and tax credits, the European Union increasingly on carbon pricing. The instrument closest to our setting is the forthcoming carbon price for buildings and road transport (ETS2), which raises the retail price of household fossil energy by design (European Commission, 2026): the same brown price our crisis raises by accident, and the object our ex-ante experiment moves (Section 7). Dietrich et al. (2025) are the closest antecedent on the monetary side: in a representative-agent model with brown and green durables, a carbon-pricing transition is mildly inflationary and green-durable purchases are markedly interest-rate sensitive, so a strict inflation target slows decarbonisation. We keep that price-stability channel and add what a representative agent cannot carry: household heterogeneity, an energy shock, and the fiscal and open-economy blocks that let us rank crisis instruments.

A distributional strand asks who bears the cost of climate policy and how revenue recycling reshapes it (Fried et al., 2018, 2024; Ascari et al., 2025); Klenert et al. (2018) review the recycling design that makes carbon pricing politically durable. Closest to us are the papers that let households choose the technology. Kuhn and Schlattmann (2024) study green subsidies when households replace a polluting old durable with a clean new one and find that subsidies should target high-income households; Crucitti et al. (2024) place heat-pump adoption in a life-cycle model with liquidity constraints and show that financial frictions are the binding barrier, so installation subsidies favour poorer households while electricity-tax cuts favour richer ones; Fried et al. (2025) build emissions from home-production tasks with clean and dirty equipment and heterogeneous distaste for emissions, and find that subsidies should target equipment at intermediate adoption and be withdrawn from the highest incomes. On the firm side, Lanteri and Rampini (2025) model extensive-margin investment in cleaner capital under credit constraints.

These papers adopt a long-run, transition perspective and study adoption in the absence of a macroeconomic disturbance; with the exception of Fried et al. (2025) they abstract from general equilibrium, and none carries monetary policy. We embed the household adoption margin in a full HANK economy hit by an energy shock, and study how carbon pricing, green subsidies, transfers and the monetary stance jointly move the pace of adoption and the transmission of the shock. The question shifts from how a policy-driven transition unfolds in calm times to how a crisis, and the instruments that meet it, trade off cushioning the short-run cost of the shock against preserving the adjustment that reduces future exposure to it.

2.3 Durable adoption and its identification

The adoption margin is a discrete durable choice inside an incomplete-markets household. We solve it with the discrete–continuous endogenous grid method of Iskhakov et al. (2017), whose extreme-value taste shocks smooth the choice into differentiable Jacobians for the sequence-space solution. In our calibration those taste shocks are not only a numerical device: at a rare adoption margin the logit scale is the structural semi-elasticity of switching (Appendix D), so it must be identified, not merely set small.

We discipline it against the quasi-experimental literature on durable adoption. Gallagher and Muehlegger (2011) and Chandra et al. (2010) estimate that a \$1,000 tax incentive raises hybrid-vehicle adoption by 31–38% among higher-income early adopters; Muehlegger and Rapson (2022) estimate a demand elasticity of -2.1 for electric vehicles among low- and middle-income households, roughly 8% per \$1,000 at their mean transaction price, the mass-market floor on responsiveness. Beresteanu and Li (2011) document the operating-cost margin, gasoline prices and hybrid demand, the analogue of our energy-cost channel. These are estimates of the upfront- and operating-cost elasticities the model needs, and Section 4 maps them onto the taste scale. The structural discrete-choice tradition that estimates such scales from rare, lumpy adjustment (Rust, 1987) is the right reference point for our regime, in contrast to the interior-probability margins, labour-force participation, where the smoothing reading applies and the scale is a numerical convenience.

3 The model

The economy is a small open economy that imports its fossil energy. It is populated by heterogeneous households, competitive firms producing a domestic good from labour, a union that sets wages, a government, and a rest of the world that sets world prices. With one exception, the environment is the small open economy HANK of Auclert et al. (2024): households consume a CES bundle of energy and goods, import prices are sticky, monopoly profits are capitalised into a mutual fund, and monetary policy holds the real interest rate constant.

The exception is the household's energy technology. In Auclert et al. (2024) every household buys energy at the same price with a fixed technology. Here each household operates a discrete durable, brown or green, a gas boiler or a heat pump, a combustion or an electric vehicle, and can pay a one-time cost ψ_g to switch from brown to green, after which it buys energy at the green price and is insulated from the fossil price. This single change ramifies in three places, and only three: the household block acquires a discrete technology state and an adoption choice; the price system acquires a second household energy price and a technology-specific cost of living; and the balance of payments acquires two flows, the switching cost and the adopters' energy saving. Everything else is inherited.

Section 3.1 presents the household block, where the new economics lives. Sections 3.2–3.5 lay out the production side, the price system, the open-economy accounts, and the government, which are standard. Section 3.6 defines equilibrium, and Section 3.7 the steady state and the two crisis experiments. Time is discrete and quarterly. A bar denotes a steady-state value; x_{-1} and x_{+1} a lag and an expected lead. All retail prices are expressed relative to the CPI P and carry the subscript $_{/P}$.

3.1 Households

There are three permanent discount-factor types $i \in \{0, 1, 2\}$ of equal mass, with discount factors β_i chosen in Section 4 to match aggregate MPCs. A household is described by three state variables: its assets a, its idiosyncratic labour productivity e, and the durable technology $d \in \{B, G\}$ it operates. Productivity follows a discretised log-AR(1), $\log e' = \rho_e \log e + \epsilon$, with a mean-one grid and transition matrix Π, as in Auclert et al. (2024). The technology is the new object, and because the paper concerns a decision over it, we begin by fixing precisely when, within a period, the household holds it, loses it, and chooses it.

3.1.1 The technology and the timing of the choice

A household enters the period operating the technology it chose last period, which we denote d_{-1}. Three events then occur, in this order, before it consumes.

1. *Breakdown.* A green durable fails with probability δ_g and reverts to brown; a brown durable is unaffected. We write d_- for the technology the household holds *after* this draw: the technology it brings to the decision. Hence $d_- = B$ whenever $d_{-1} = B$, while $d_- = G$ with probability $1 - \delta_g$ when $d_{-1} = G$.

2. *Productivity.* The household draws its labour productivity e.

3. *The decision.* The household chooses, in a single problem, the technology d it will operate together with its consumption and saving. A household holding brown ($d_- = B$) may keep brown or pay the switching cost ψ_g to adopt green; a household holding green ($d_- = G$) operates it, since running an owned green durable costs nothing and reverting to brown is never chosen.

No information arrives between the productivity draw and the decision, so the household settles technology, consumption, and saving *jointly* (Section 3.1.3). The technology it operates is what it carries forward, d becomes next period's d_{-1}, so the within-period sequence is

$$\underbrace{d_{-1}}_{\text{held on entry}} \xrightarrow{\text{breakdown}} \underbrace{d_-}_{\text{held at the decision}} \xrightarrow{\text{choice}} \underbrace{d}_{\text{operated, carried forward}}.$$

Two binary objects thus describe the technology: the holding at the decision, d_-, and the operated technology, d. The switching cost is paid exactly when a brown holder adopts, that is when $d_- = B$ and $d = G$; we write $\mathbb{I}[d_- = B, d = G]$ for a switcher and $\mathbb{I}[d = G]$ for a household operating green. The feasible choices are

$$\mathcal{F}(d_-) = \begin{cases} \{B,\, G\} & d_- = B \quad \text{(keep brown, or pay } \psi_g \text{ to adopt)}, \\ \{G\} & d_- = G \quad \text{(operate the green durable)}. \end{cases}$$

Two features follow directly, and both recur below. First, the technology is persistent but asymmetric: brown is absorbing, and green is lost only through breakdown. Second, the adoption decision is, in effect, an option held by the brown population alone: a green household has nothing to choose.

A constant green share therefore requires a standing flow of switchers to replace the durables that fail. In steady state this flow is

$$\bar{D}^{\mathrm{sw}} = \delta_g\, \bar{D^G}, \tag{1}$$

with D^G and D^{sw} the population shares of green households and of switchers. At the calibrated $\bar{D^G} = 5\%$ and a five-year durable life, this flow is about a quarter of one percent of households per quarter: adoption is a rare, lumpy event, which is what makes its elasticity, not its level, the object to identify (Section 4.1, Appendix D).

3.1.2 Preferences and the cost of living

Labour supply is determined at the union level (Section 3.2), so flow utility is CRRA over a consumption bundle c alone,

$$u(c) = \frac{c^{\,1-\sigma}}{1-\sigma}, \qquad u(c) = \log c \ \text{ for } \sigma = 1, \tag{2}$$

with σ the inverse elasticity of intertemporal substitution, as in Auclert et al. (2024). The bundle is the Auclert et al. (2024) CES nest of energy against a home/foreign goods composite, with energy weight α_E and a low substitution elasticity η_E.

The departure from AMRS is that energy comes at a technology-specific price: a household operating brown buys energy at P^E_B, one operating green at $P^E_G < P^E_B$. The operated technology therefore fixes both the energy price and the exact cost-of-living index,

$$p^E_d = P^E_B + \mathbb{I}[d = G]\left(P^E_G - P^E_B\right), \tag{3}$$

$$\frac{p_G}{p_B} = \left[\frac{(1-\alpha_E)\, p_{HF/P}^{1-\eta_E} + \alpha_E\,(P^E_G)^{1-\eta_E}}{(1-\alpha_E)\, p_{HF/P}^{1-\eta_E} + \alpha_E\,(P^E_B)^{1-\eta_E}}\right]^{\frac{1}{1-\eta_E}}, \qquad \frac{p_G}{p_B} = \left(P^E_G / P^E_B\right)^{\alpha_E} \ \text{ for } \eta_E = 1. \tag{4}$$

The adoption choice responds to the relative saving $p_G/p_B < 1$: the cheaper energy an adopter enjoys, the object the switching cost buys. It is the ratio of the two technology bundles (20), invariant to how the CPI is normalised; at the calibrated steady state $p_G/p_B \approx 0.992$.

3.1.3 The household's problem

A household of type i arrives at the consumption stage with assets a and productivity e, operating the technology d it has just chosen, having held d_- at the decision. Before writing the program it is worth stating, in words, what the household has and what it does with it. On the resource side there are three items. Assets earn the ex-post real return r, the return on the mutual fund of Section 3.2; it coincides with the lagged ex-ante rate r_{-1}^{ante} at every date except the first, when the date-0 revaluation of the fund's equity drives the two apart (equation (13)), and the household bears that surprise. Labour income is $e\,Z$, where $Z \equiv (1-\tau^L)wn$ is aggregate after-tax labour income distributed in proportion to productivity, as in Auclert et al. (2024); the household does not choose hours, which the union sets (Section 3.2). The government may pay a transfer T, specified below. On the spending side, the household buys the consumption bundle at its technology-specific cost of living p_d, saves $a' \geq 0$ subject to the borrowing constraint, and, if it is a brown holder adopting green this period, pays the switching cost ψ_g, once. The budget constraint collects these items,[1]

$$p_d\, c + a' + \psi_g\, \mathbb{I}[d_- = B,\, d = G] = (1+r)\,a + e\,Z + T. \tag{5}$$

The switching cost is the economics of the margin in one term. It is paid exactly once, in the period a brown holder adopts, and it buys the permanently cheaper bundle $p_G < 1$ for as long as the durable survives: an upfront, lumpy outlay set against a gradual operating saving. Because the outlay is financed out of current resources against the borrowing constraint, wealth governs who can undertake it, which is why adoption concentrates among the unconstrained (Section 4.2).

The transfer carries ex-post fiscal policy,

$$T = T^{\text{unt}} + ins_E\,(p^E - \bar{p}^E)\,\bar{c}_{E,i}^{B}(e,a), \tag{6}$$

an untargeted lump sum T^{unt} plus a targeted (Slutsky) transfer that scales, with insurance proportion ins_E, the crisis rise in the energy price by the household's *pre-crisis* brown-energy demand $\bar{c}_{E,i}^{B}(e,a)$; both are zero in the baseline. One design feature matters throughout: the

[1] The unit of account is the CPI basket (Section 3.6), so $p^{\text{num}} = 1$: asset returns, wages, transfers and ψ_g in (5) are already in CPI units and enter without conversion.

transfer conditions on steady-state energy use, never on the current technology. A household that switches forfeits nothing, so the transfer leaves the adoption margin undistorted, in sharp contrast to the price cap of Section 3.3, which acts on P_B^E itself.

The household chooses its technology, consumption, and saving *together*, in a single problem:

$$\max_{d \in \mathcal{F}(d_-)} \; \max_{c,\, a' \geq 0} \left\{ u(c) + \beta_i \, \mathbb{E}\Big[\bar{V}(d'_-, e', a')\Big] \right\} \quad \text{subject to (5)}, \tag{7}$$

with $\bar{V}$ the continuation value (defined in (10)) and next period's holding d'_- given by breakdown applied to the operated technology d. The nested maxima are not a temporal sequence, since nothing is realised between them, but the structure of a discrete–continuous choice: for each feasible technology the household solves the continuous consumption-saving problem, obtaining the technology-conditional value $V(d \mid d_-, e, a)$, in which the origin d_- enters only through the switching cost; it then selects the technology (Section 3.1.4). Only three continuous problems are ever relevant: a brown household (p_B, no switching cost), a green incumbent (p_G, no switching cost), and a green adopter (p_G, resources reduced by ψ_g). Each satisfies the usual Euler and envelope conditions,

$$u'(c) = \beta_i \, p_d \, \mathbb{E}\Big[\bar{V}_a(d'_-, e', a')\Big], \qquad V_a = (1 + r)\, u'(c)/p_d, \tag{8}$$

with $u'(c) = c^{-\sigma}$; a green household's cheaper bundle raises the consumption value of every unit of wealth. CES demands split by the operated technology,

$$c_E(d) = \alpha_E \Big(p_d^E / p_d\Big)^{-\eta_E} c, \qquad c_{HF}(d) = (1 - \alpha_E)\Big(p_{HF/P}/p_d\Big)^{-\eta_E} c, \tag{9}$$

and brown versus green energy use follow the technology, $c_E^B = c_E\,(1 - \mathbb{I}[d = G])$ and $c_E^G = c_E \, \mathbb{I}[d = G]$.

3.1.4 The technology choice

The outer maximum in (7) selects the technology, completing the joint problem. A household holding d_- compares the technology-conditional values $V(d \mid d_-, \cdot)$; with an i.i.d. extreme-value taste shock of scale σ_ε on each option, the choice is logit and the household's continuation value $\bar{V}$ is the log-sum-exp,

$$\begin{aligned} \Pr(d \mid d_-, e, a) &= \frac{\exp\{V(d \mid d_-, e, a)/\sigma_\varepsilon\}}{\sum_{d' \in \mathcal{F}(d_-)} \exp\{V(d' \mid d_-, e, a)/\sigma_\varepsilon\}}, \\ \bar{V}(d_-, e, a) &= \sigma_\varepsilon \log \sum_{d' \in \mathcal{F}(d_-)} \exp\Big(V(d' \mid d_-, e, a)/\sigma_\varepsilon\Big). \end{aligned} \tag{10}$$

For a green holder the feasible set is a singleton, so $\bar{V}(G,\cdot) = V(G \mid G,\cdot)$ and there is nothing to decide. The only genuine choice is the brown holder's: a two-way comparison of staying brown against paying ψ_g to adopt green. That cost already sits inside the adopter's budget constraint (5), so the option values price it automatically.

The economics of the margin is transparent in steady state. The marginal switcher equates the premium to the discounted operating-cost saving over the durable's expected life,

$$\psi_g \;\approx\; \frac{\left(\bar{P}^E_B - \bar{P}^E_G\right)\bar{C}^G_E}{r+\delta_g}, \tag{11}$$

which ties the adoption threshold to the delivered cost ratio $\varrho = \bar{P}^E_G/\bar{P}^E_B$. Every experiment in the paper acts on this one comparison. An energy crisis raises P^E_B and lifts the return to switching; a retail price cap holds P^E_B at its pre-crisis level and removes that return entirely; a transfer supports income while leaving P^E_B, and hence (11), free; a permanent carbon tax raises the brown price in steady state, greening the economy before any crisis arrives.

Away from steady state, the responsiveness of the margin is governed by σ_ε. The taste shocks are, in the tradition of Iskhakov et al. (2017), the smoothing that makes the discrete choice differentiable; at a rare margin they are also the structural semi-elasticity of switching, so σ_ε cannot be treated as a nuisance parameter and set small. Section 4.1 identifies it from quasi-experimental adoption elasticities, and Appendix D develops the smoothing–elasticity duality.

3.1.5 Aggregation

For each type i, aggregates integrate the household policies over the stationary distribution Λ_i, and the economy averages the three types equally,

$$X = \tfrac{1}{3}\textstyle\sum_i X_i, \qquad X \in \{C,\ A,\ D^G,\ D^{\mathrm{sw}},\ C^B_E,\ C^G_E,\ C_{HF},\dots\}. \tag{12}$$

Aggregate energy demand is $c_E = C^B_E + C^G_E$, and home- and foreign-good demands apply the common inner nest to the composite, $c_H = (1-\alpha_F)\,p_{H/HF}^{-\eta}\,C_{HF}$ and $c_F = \alpha_F\,p_{F/HF}^{-\eta}\,C_{HF}$. The two aggregates the policy experiments track are the green share D^G, the population share operating green, and the switching flow D^{sw}, the share adopting green each period.

3.2 Production and wage setting

Production and profits. Competitive firms produce the domestic final good from labour with productivity A^N, sold at a constant gross markup μ, so output is $y = A^N n$ and after-tax aggregate labour income is $Z = (1-\tau^L)wn$, with n hours and w the real wage. Monopoly profits are capitalised into a mutual fund that households hold,

$$D = (\mu - 1)\, wn, \qquad J = D + \frac{J_{+1}}{1 + r^{\text{ante}}}, \qquad j = \frac{J_{+1}}{1 + r^{\text{ante}}}, \qquad 1 + r = \frac{J}{j_{-1}}, \tag{13}$$

following the notation of Auclert et al. (2024): r_t^{ante} is the ex-ante real rate at which the risk-neutral fund prices its claims (equation (27)), j is the ex-dividend share price, and r is the fund's ex-post return: the return households earn in (5). Because the fund equates ex-ante returns across assets, the ex-post return equals the lagged ex-ante rate at every date but the first, $r_t = r_{t-1}^{\text{ante}}$ for $t \geq 1$; only the impact return r_0 differs, reflecting the date-0 revaluation of the fund's equity, which is the source of the revaluation term in (26). Importers are foreign-owned and the energy endowment is not held domestically, so domestic firms are the only domestically held risky claim.

Wage setting. A union sets the nominal wage under Calvo frictions θ_w, trading off households' marginal rate of substitution against the after-tax wage, with a real-wage-stabilisation term (weight ζ_{BG}) as in Auclert et al. (2024). With slope $\kappa_w = (1-\theta_w)(1-\beta\theta_w)/\theta_w$, inverse Frisch elasticity φ, and labour-disutility constant v_φ, wage inflation follows

$$\pi^w(1+\pi^w) = \beta\, \pi^w_{+1}(1+\pi^w_{+1}) + \frac{\kappa_w}{v_\varphi\, n^\varphi}\left[v_\varphi\, n^\varphi - \frac{(1-\tau^L)w\, C^{-\sigma}}{\mu} - \zeta_{BG}\frac{(w-\bar{w})\, w\, \bar{C}^{-\sigma}\bar{n}}{\mu\, n\, \bar{w}} \right], \tag{14}$$

and the real wage evolves as $w = (1+\pi^w)\, w_{-1}/(1+\pi)$.

3.3 The price system

The paper's mechanism runs entirely through relative prices, so it is worth laying the price system out in one place. Table 1 collects it: world prices at the top, a wholesale layer where nominal rigidities live, a retail layer where policy operates, and the CPI that aggregates them.

Symbol	Object	Determined by
P^{E*}, P_F^*, C^*, r^*	world energy and goods prices, demand, rate	exogenous[a]
p^E	wholesale retail price of brown energy	NKPC (17)
P_B^E	brown households' energy price	cap rule (18)
P_G^E	green households' energy price	$\varrho\,\bar{p}^E$, eq. (19)
$p_{H/P}$	domestic good	markup pricing (15)
$p_{F/P}$	imported consumption good	NKPC (16)
$p_{HF/P}$, $p_{H/HF}$	goods composites	outer nest (21)
p_B, p_G	brown/green cost of living	bundle nests (20); saving p_G/p_B (4)
Q	real exchange rate	UIP (22)

Table 1: The price system. All retail prices are relative to the CPI. [a]Under the fixed-quantity closure the world energy price P^{E*} clears the energy market (Section 3.6).

The domestic good. Marginal-cost pricing at the constant markup gives

$$p_{H/P} = \frac{\mu}{A^N}\, w. \tag{15}$$

Imported goods and wholesale energy. Retailers of the imported good and of energy reset prices under Calvo frictions θ_F and θ_E, so world price movements pass through to domestic retail prices only gradually. With the usual slopes κ_F and κ_E,

$$\kappa_F\Big(\frac{QP_F^*}{p_{F/P}} - 1\Big) + \beta_F\,\pi_{+1}^F(1+\pi_{+1}^F) - \pi^F(1+\pi^F) = 0, \tag{16}$$

$$\kappa_E\Big(\frac{QP^{E*}}{p^E} - 1\Big) + \beta_E\,\pi_{+1}^E(1+\pi_{+1}^E) - \pi^E(1+\pi^E) = 0. \tag{17}$$

The margins these retailers earn out of steady state accrue abroad (the importers are foreign-owned, as in Auclert et al., 2024); they matter below only because the import bill is valued at retail rather than world prices.

The retail energy prices, where policy lives. The price a brown household actually pays applies the cap to the wholesale price,

$$P_B^E = (1-\tau^E)\,p^E + \tau^E\,\bar{p}^E, \tag{18}$$

so τ^E dials pass-through continuously: at $\tau^E = 0$ households face the wholesale price, at $\tau^E = 1$ the brown retail price is frozen at its pre-crisis level: the full shield. The green price is fixed relative to the pre-crisis brown price,

$$P_G^E = \varrho\,\bar{p}^E, \qquad \varrho < 1, \tag{19}$$

so adopters are fully insulated from the fossil shock.

The consumer price index. Each technology has its own consumption bundle. Writing p_B, p_G for the brown and green bundle prices relative to the CPI, the inner (energy versus goods) CES nests are

$$p_d^{1-\eta_E} = (1-\alpha_E)\,p_{HF/P}^{1-\eta_E} + \alpha_E\,(P_d^E)^{1-\eta_E}, \qquad d \in \{B, G\}, \tag{20}$$

with the outer (home versus foreign) nest

$$(1-\alpha_F)\,p_{H/HF}^{1-\eta} + \alpha_F\,p_{F/HF}^{1-\eta} = 1. \tag{21}$$

The CPI is the expenditure-weighted index of the two bundles: P is defined by the requirement that nominal consumption expenditure deflated by P equal real consumption, $\int p_{d(i)}\,c(i)\,\mathrm{d}i = \int c(i)\,\mathrm{d}i$, so that $\sum_d \omega_d\,p_d = 1$ with ω_d the consumption share of state d. As green diffuses the energy leg is reweighted from brown toward green through $\omega_G = D^G$, so P is the true cost-of-living index the monetary rule reads; no measurement gap opens between the model's price index and the cost of living.

3.4 The open economy

Foreign demand and the exchange rate. Foreign demand for the home good is $c_H^* = \alpha^*(p_H^*)^{-\gamma}C^*$, with trade elasticity γ, and the real exchange rate is pinned by uncovered interest parity,

$$\frac{Q}{Q_{+1}}\left(1 + r^{\text{ante}}\right) = 1 + r^*. \tag{22}$$

World energy supply. World supply is a fixed endowment E^* plus intertemporal stock arbitrage with cost Γ,

$$E^{\text{stock}} = \frac{P_{+1}^{E*}/(1+r^*) - P^{E*}}{\Gamma}, \qquad E^s = E^* + \left(E_{-1}^{\text{stock}} - E^{\text{stock}}\right). \tag{23}$$

The home economy owns none of the endowment, it is a pure importer, so energy rents never enter domestic wealth.

The balance of payments and the adoption flows. The adoption margin adds exactly two terms to the current account, and they pull in opposite directions. The switching cost is a resource outflow, booked as an import M^{sw}: it is front-loaded, paid once per switcher, and scales with the switching volume. The adopters' saved fossil energy is a resource inflow, booked as an import saving S^E: it accrues gradually, over the life of each green durable,

$$M^{\text{sw}} = \psi_g \, D^{\text{sw}}, \qquad S^E = (P_B^E - P_G^E) \, C_E^G. \tag{24}$$

Net exports collect trade in goods and energy together with these two flows,

$$nx = Q \, p_H^* c_H^* - p_{F/P} \, c_F - p^E \, c_E - M^{\text{sw}} + S^E, \tag{25}$$

and net foreign assets accumulate at the ex-ante rate, with one additional term,

$$nfa = nx + (r - r_{-1}^{\text{ante}})\Big(A_{-1}^{\text{CPI}} - j_{-1}\Big) + (1 + r_{-1}^{\text{ante}}) \, nfa_{-1}. \tag{26}$$

The middle term is a revaluation, and it is worth unpacking. Households earn the ex-post return r on their wealth, while the country's accounts accrue at the promised ex-ante rate r_{-1}^{ante}; the two differ only at date 0, when the shock revalues the equity of the mutual fund (equation (13)). Applying the surprise $r - r_{-1}^{\text{ante}}$ to households' wealth net of the domestically held fund, $A_{-1}^{\text{CPI}} - j_{-1}$, records the one-off capital gain or loss on the country's external claims. At date 0, the only date the term is active, $A_{-1}^{\text{CPI}} - j_{-1}$ equals the net foreign position itself, since government debt is zero at the pre-crisis steady state; from date 1 onward $r_t = r_{t-1}^{\text{ante}}$ and the term vanishes.

Booking the switching cost and the green energy as imports is a convention, not a result: it fixes the sign of the adoption channel's *output* contribution but none of the policy rankings, which run through relative prices alone. The alternative domestic booking is collected in Appendix A.

3.5 The government

Monetary policy. The nominal rate follows a rule in current and expected inflation, with the real rate defined by the Fisher equation,

$$1 + i = (1 + \bar{r}^*)(1 + \phi_\pi \pi)(1 + \phi_{\pi e} \pi_{+1}) + \varepsilon^m, \qquad r^{\text{ante}} = \frac{1 + i}{1 + \pi_{+1}} - 1, \tag{27}$$

where ε^m is a monetary shock, zero outside the monetary experiment of Appendix C. The baseline is the constant-real-rate rule of Auclert et al. (2024), $(\phi_\pi, \phi_{\pi e}) = (0, 1)$. Through

(22) it pins the path of the real exchange rate, so a domestic transfer has no terms-of-trade effect: a property inherited from AMRS, not assumed anew. A standard Taylor rule, $(\phi_\pi, \phi_{\pi e}) = (1.5, 0)$, is reported as a robustness contrast in Appendix C.

Fiscal policy. Fiscal policy is the object of Sections 6–7, so we lay out each instrument explicitly. Four are crisis instruments, deployed once the shock hits and zero in the baseline; a fifth, the carbon tax, is permanent and defines the "prepared" economy of Section 7. Table 2 states, for each, what it acts on, the base its cost scales with, and its budgetary cost.

Instrument	Acts on	Cost scales with	Fiscal cost
Price cap (τ^E)	brown retail price P_B^E	actual energy use	$\tau^E(p^E - \bar{p}^E)\, c_E$
Slutsky transfer (ins_E)	household income	pre-crisis brown use	$ins_E\,(p^E - \bar{p}^E)\,\bar{C}_E^B$
Flat transfer (T^{unt})	household income	none (lump sum)	T^{unt}
Green subsidy (s_g)	switching cost ψ_g	number of switchers	$s_g\,\psi_g\,D^{\text{sw}}$
Carbon tax (τ_b)	brown price P_B^E, permanent	brown energy use	rebated (App. A)

Table 2: Fiscal instruments. The first four are crisis instruments, zero in the baseline (Sections 6–6.6); the carbon tax is permanent and defines the prepared economy (Section 7). Its revenue is rebated lump sum in the baseline, with green-subsidy recycling in Appendix A.

The distinction the results turn on is between acting on the *price* and acting on *income*. The price cap holds P_B^E down through (18); because P_B^E is precisely what the adoption margin responds to (equation (11)), capping the price switches the margin off, and the cap's cost is a marginal subsidy: it scales with actual crisis energy use. The two transfers instead leave P_B^E free and support income, so the margin survives. They differ in their base: the *Slutsky* transfer (intensity ins_E) is indexed to each household's pre-crisis brown-energy use, so a switcher keeps the same cheque and the margin is undistorted, but, as Section 6.6 shows, indexing to energy use is regressive; the *flat* transfer T^{unt} is the equal-cost, unconditional alternative. The *green subsidy* pays a fraction s_g of the switching cost, forcing adoption directly rather than through the price. The *carbon tax* raises P_B^E permanently, greening the economy before any crisis; it is the ex-ante counterpart to the crisis instruments and the subject of Section 7.

All crisis outlays are deficit-financed. Collecting them term by term, with each the cost line of Table 2,

$$X = \underbrace{T^{\text{unt}}}_{\text{flat transfer}} + \underbrace{ins_E\,(p^E - \bar{p}^E)\,\bar{C}_E^B}_{\text{Slutsky transfer}} + \underbrace{\tau^E\,(p^E - \bar{p}^E)\,c_E}_{\text{price cap}} + \underbrace{s_g\,\psi_g\,D^{\text{sw}}}_{\text{green subsidy}}, \tag{28}$$

debt is stabilised by the distortionary labour tax,

$$B = (1 + r_{-1}^{\text{ante}})B_{-1} + X - \tau^L wn, \qquad \tau^L = \psi_B\,(B_{-1} - \bar{B}), \tag{29}$$

with no debt in steady state ($\bar{B} = 0$): crisis spending is repaid by raising the labour tax in proportion to the debt it creates.

3.6 Equilibrium

Numeraire. The unit of account is the CPI basket, the AMRS convention: $p^{\text{num}} = 1$ and the household's budget is stated directly in real terms. Because this CPI is the true expenditure deflator, real allocations, relative prices and the market-clearing conditions are all expressed in real terms and are independent of the unit of account: adoption reweights the basket, and hence the price level P, but leaves every real object unchanged. The CPI plays only its two economic roles, deflating the household budget and serving as the index the monetary rule targets, with no separate accounting wedge.

Market clearing. Goods, assets, and energy clear:

$$c_H + c_H^* - y = 0, \qquad A^{\text{CPI}} - nfa - j - B = 0, \qquad \begin{cases} P^{E*} \text{ exogenous} & \text{(elastic supply)}, \\ c_E = E^s & \text{(fixed quantity)}, \end{cases} \tag{30}$$

where the asset-market condition is redundant by Walras' law: it holds once goods, energy, and the government budget balance, so we do not impose it separately. The two energy closures correspond to the two experiments of Section 5: a price-taking economy hit by an exogenous world-price path, and a fixed world quantity whose price clears the market.

Let Ψ_t denote the joint distribution over the household state (d, e, a), durable technology, idiosyncratic productivity, and assets, at date t. Taking as given the sequences of prices, taxes, and transfers, each household solves the recursive program of Section 3.1.3: it chooses nondurable consumption and savings by (7)–(8) and, at the durable stage, switches from brown to green with the logit probability (10). This delivers value functions V_t and policy functions $\{c_t, a_t', P_t^{sw}\}(d, e, a)$. The distribution then evolves as $\Psi_{t+1} = \Lambda_t(\Psi_t)$, where the law of motion Λ_t is induced by these policies together with the exogenous productivity chain, and aggregates are the corresponding moments of Ψ_t (12).

Definition (competitive equilibrium). Given the world paths $\{P^{E*}, r^*, P^{F*}\}$, the initial distribution Ψ_0 and net foreign assets, and the monetary and fiscal rules, a competitive equilibrium is a sequence of aggregate prices $\{w, \pi, \pi^w, \pi^F, \pi^E, Q, p^E, r, r^{\text{ante}}, p_{H/P}\}_t$, value and policy functions $\{V_t, c_t, a'_t, P_t^{sw}\}$, distributions $\{\Psi_t\}$, aggregate quantities $\{y, n, c_H, c_F, c_E, C, A\}_t$, technology shares $\{D^G, D^{\text{sw}}\}_t$, and fiscal variables $\{B, \tau^L\}_t$ such that:

(i) *(Households)* given prices, taxes, and transfers, the value and policy functions solve the recursive problem (7)–(8), with brown-to-green switching governed by (10)–(11);

(ii) *(Consistency)* Ψ_t evolves from Ψ_0 under the law of motion induced by the policy functions and the productivity process, and the aggregates in (12) are the moments of Ψ_t;

(iii) *(Firms and unions)* the price and wage Phillips curves (13)–(14) and the price system (15)–(19) hold;

(iv) *(Government)* the monetary rule (27), the fiscal rule, and the government budget (29) hold;

(v) *(Open economy)* uncovered interest parity (22), the balance-of-payments and net-foreign-asset accounting (25)–(26), and the energy-supply closure (23) hold;

(vi) *(Clearing)* the goods and energy markets clear (30); the asset market clears by Walras' law.

This is the object of interest: a sequence of prices and allocations, together with a distribution consistent with individual optimisation. The sequence-space Jacobian method of Section 3.7 computes the linearised transition of this equilibrium.

3.7 Steady state and the two experiments

As a pure importer, the openness normalisations of Auclert et al. (2024) reduce to $\alpha_F = (\alpha - \alpha_E)/(1 - \alpha_E)$ and a trade elasticity $\gamma = \eta$ set to hit an import-demand target $\chi = 0.3$, with scale normalisations $y = 1$, $A^N = \mu$, $\alpha^* = \alpha$, and $\bar{E}^* = \alpha_E$. The steady state solves for the labour-disutility shifter ϕ, the top discount factor $\beta_{\max}$, output, and the switching cost ψ_g, against the wage Phillips curve, the net-foreign-asset condition, goods clearing, and the green-share target $\bar{D^G} = 5\%$. The switching cost is thus calibrated to the level of adoption, and Section 4 takes up the parameter the level cannot pin, σ_ε. The steady state is identical across all baseline experiments.

Two experiments drive the crisis, one per energy closure. The *price* shock doubles the world energy price on impact and lets it decay with a 16-quarter half-life under perfectly elastic supply,

$$P_t^{E*} = \bar{P}^{E*}\Big(1 + \text{size} \cdot \rho^t\Big), \qquad \rho = 2^{-1/\text{half-life}}, \tag{31}$$

the experiment of Auclert et al. (2024). The *supply* shock removes 10% of the world quantity for six quarters and lets the world price clear the market,

$$E_t^* = \bar{E}^* \Big(1 - \text{drop} \cdot \mathbb{I}[t < \text{quarters}]\Big), \tag{32}$$

the closure of Bayer et al. (2026). Section 5 puts the adoption margin to work under both.

4 Calibration

The aggregate block is inherited from Auclert et al. (2024); the durable/adoption block is this paper's addition. We keep the aggregate calibration close to a generic energy importer, the AMRS parameters, an energy expenditure share of four percent, a low intensive substitution elasticity, and read the durable as any household energy durable, a heat pump or an electric vehicle. The economy we speak to is the European one of Sections 1 and 8; the aggregate calibration is an illustrative mapping onto it rather than a euro-area estimation. The steady state is identical across all baseline experiments. Table 3 collects the parameters.

Parameter	Value	Source / target
Preferences and idiosyncratic risk		
r real rate (quarterly)	0.01	Auclert et al. (2024)
σ inverse EIS	1 (log)	Auclert et al. (2024)
φ inverse Frisch	2	Frisch 0.5 (standard)
σ_e, ρ_e, n_e income process	0.57, 0.92, 7	Rouwenhorst; AMRS baseline
β types / spread / $\beta_{\max}$	3 / 0.06 / solved	mean MPC ≈ 0.30; $\beta_{\max}$ closes nfa
asset grid $[\underline{a}, \bar{a}]$, n_a	$[0, 400]$, 50	
Consumption nest and openness		
α_E energy expenditure share	0.04	Auclert et al. (2024)
η_E energy substitution	0.10	Auclert et al. (2024)
α total import share	0.30	Auclert et al. (2024)
$\alpha_F,\ \gamma = \eta$	derived	import-demand target $\chi = 0.3$
Firms, prices, wages		
μ gross markup	1.03	Auclert et al. (2024)
θ_w wage Calvo	0.938	AMRS baseline
θ_E energy-price Calvo	0.65	AMRS baseline
θ_F import-price Calvo	0.90	AMRS baseline
ζ_{BG} real-wage stabilisation	5	AMRS baseline
Energy supply and open economy		
ε^E energy-supply closure	∞ / fixed $\bar{E}$	price-taking SOE / fixed-quantity (Bayer et al. (2026))
Γ stock arbitrage	100	AMRS baseline
$\bar{E}^{\text{shock}}$ world supply	α_E	endowment normalisation
Monetary and fiscal		
$(\phi_\pi, \phi_{\pi e})$ baseline / Taylor	$(0, 1)$ / $(1.5, 0)$	constant real rate (AMRS) / Taylor
$\bar{B},\ \psi_B$ debt / feedback	0, 0.04	no SS debt; τ^Y solved
Green adoption margin [this paper]		
δ_g green breakdown rate	0.05	five-year durable life
$\varrho = \bar{P}^E_G / \bar{P}^E_B$	0.54	EV/ICE operating-cost ratio; App. E
ψ_g switching cost	0.770 (solved)	targets $\bar{D}^G = 0.05$
σ_ε logit taste scale	0.05	identified in Section 4.1
Policy instruments (all 0 in the baseline)		
τ^E price-cap intensity	$\{0, \ldots, 1\}$	1 = full freeze
ins_E targeted transfer	$\{0, 1\}$	pre-crisis brown-energy indexed
T^{unt} untargeted transfer	$\{0, 1\}$	flat-transfer counterpart
s_g switching subsidy	$\{0, 1\}$	crisis / carbon-financed
ETS (prepared economy only)		
τ_b brown carbon tax	$\{0.10, 0.20, 0.30\}$	permanent, ψ_g fixed
τ_g green carbon tax	≈ 0	green near-clean
revenue recycling	rebate / green subsidy	Appendix A

Table 3: Calibration. The aggregate block is inherited from Auclert et al. (2024); the durable/adoption block is this paper's addition. ψ_g is solved to $\bar{D}^G = 5\%$; σ_ε is identified in Section 4.1.

The adoption block adds four parameters. The green durable depreciates at $\delta_g = 0.05$, a five-year life, which fixes the steady-state switching flow at $\bar{D}^{\text{sw}} = \delta_g \bar{D}^G$. We anchor the durable to the electric-vehicle margin, where the operating-cost gap is measured directly. The delivered green/brown ratio is the electricity cost of driving a battery-electric vehicle relative to the fuel cost of a comparable gasoline car, $\varrho = (P^G e_G)/(P^B e_B)$. At 2025 euro-area retail prices (electricity 0.29 EUR/kWh and gasoline 1.62 EUR/L, both inclusive of taxes) and real-world energy use of 0.21 kWh/km for the electric car and 7.0 L/100km for the gasoline car, this gives $\varrho = 0.54$. Tax-inclusive prices are the relevant ones here: the baseline carries no carbon tax, so the household compares retail prices, and the asymmetric taxation of the two fuels is part of the environment the crisis and the ETS act on rather than something assumed away. Appendix E lists every input and its source, and shows the policy ranking is invariant to ϱ across $[0.48, 0.80]$. The switching cost ψ_g is solved to hit a steady-state green share $\bar{D}^G = 5\%$. That leaves one free parameter, the logit taste scale σ_ε, which governs how responsive the margin is and which a level target cannot pin. Section 4.1 disciplines it.

4.1 Identifying the taste scale

The switching cost ψ_g and the taste scale σ_ε are not separately identified by the single steady-state target $\bar{D}^G = 5\%$. For any σ_ε, ψ_g re-solves to hit the same share, so the steady state is invariant along the $(\sigma_\varepsilon, \psi_g)$ locus (Table 15): the green share and the switcher mass do not move as σ_ε ranges over an order of magnitude and ψ_g with it, from 0.32 to 1.53.

What the locus does not leave invariant is the responsiveness of the margin. The steady-state adoption elasticity, the response of the green share to a permanent improvement in the green/brown operating-cost ratio, at fixed ψ_g, falls from 2.8 at $\sigma_\varepsilon = 0.02$ to 0.4 at 0.20, and the crisis adoption wave falls with it, from a peak ΔD^G of 19.9 to 2.0 percentage points (Figure 27, in Appendix D). The adoption-channel magnitudes are therefore conditional on σ_ε: a level moment cannot discipline them, and a second moment, the adoption elasticity, is required.

The quasi-experimental vehicle literature provides exactly that moment, on the upfront-cost margin. Gallagher and Muehlegger (2011) and Chandra et al. (2010) estimate that a \$1,000 tax incentive raises hybrid adoption by 31–38%, in a population of higher-income early adopters; Muehlegger and Rapson (2022) estimate a demand elasticity of -2.1 for electric vehicles among low- and middle-income households, roughly 8% per \$1,000 at their mean transaction price of \$26,000: the mass-market floor on responsiveness. In the model one unit is quarterly household consumption, taken as \$20,000, so the baseline $\psi_g = 0.538$

is a green premium of about \$10,800; a \$1,000 cut in ψ_g raises the switching flow by 40% at $\sigma_\varepsilon = 0.05$, 31% at 0.07, and 12% at 0.20 (Table 15).

Which end of the empirical range is relevant is a statement about the marginal adopter, not a free choice. At a 5% green share the model's marginal adopters consume roughly twice the average brown household (Section 7): they are the higher-income early adopters of the hybrid-era estimates, not the mass-market population of Muehlegger and Rapson (2022), whose estimate therefore bounds responsiveness from below. Matching the early-adopter range pins $\sigma_\varepsilon \in [0.05, 0.07]$, with an implied green premium of \$11,000–\$14,000 squarely in the heat-pump and EV range, whereas $\sigma_\varepsilon \geq 0.10$ would imply premia of \$18,000 and above. The baseline $\sigma_\varepsilon = 0.05$ is the elastic end of that interval, and the mass-market estimate is spanned by the sensitivity analysis. The operating-cost margin, the channel Beresteanu and Li (2011) document for gasoline prices and hybrid demand, then implies an elasticity of the green share to the energy-cost gap of 1.3 at the baseline. This responsiveness is a structural property of a rare margin, not an artefact of the smoothing method; Appendix D develops the point.

The dollar mapping is the one assumption stated rather than estimated, and its role is transparent: a higher consumption unit scales the model's per-\$1,000 response down and the implied premium up, so the two criteria co-move and the joint interval stays tight, at \$16,000 the pin is roughly $[0.07, 0.10]$, at \$24,000 $[0.04, 0.06]$. Each of these intervals lies inside the sensitivity grid of Table 15, over which the steady state and the policy ordering are invariant.

The aggregate results do not turn on any of this. The cumulative output loss $\sum y$ is flat across the locus (Table 15, last column), varying by under 4% while the adoption response varies tenfold, and the policy ordering follows from the relative price alone. The taste scale governs how large the adoption channel is, not whether the cap switches it off or the transfer preserves it.

4.2 The steady state: who adopts

At the calibrated steady state green durables are 5% of the stock and the quarterly switching flow is 0.26%, the flow that replaces green depreciation (Table 14, Appendix D). Adoption is far from uniform across households. Figure 1 plots the switching probability of the median discount-factor type over the wealth–productivity grid: the choice *policy function*, not the mass-weighted pool, which mixes three types with very different switching rates and is best read as the distribution in Table 14. The probability that a brown incumbent switches to green rises monotonically in both wealth and idiosyncratic productivity. Wealthier households can finance the switching cost ψ_g and more productive households expect the higher future income that makes the durable worth replacing, so adoption concentrates among wealthy, high-productivity households. The surface also makes the rare-margin regime visible: away from that group the probability sits far below one: the brown mass occupies the exponential tail of the logit, where the semi-elasticity of switching is $1/\sigma_\varepsilon$ household by household (Appendix D, Table 14).

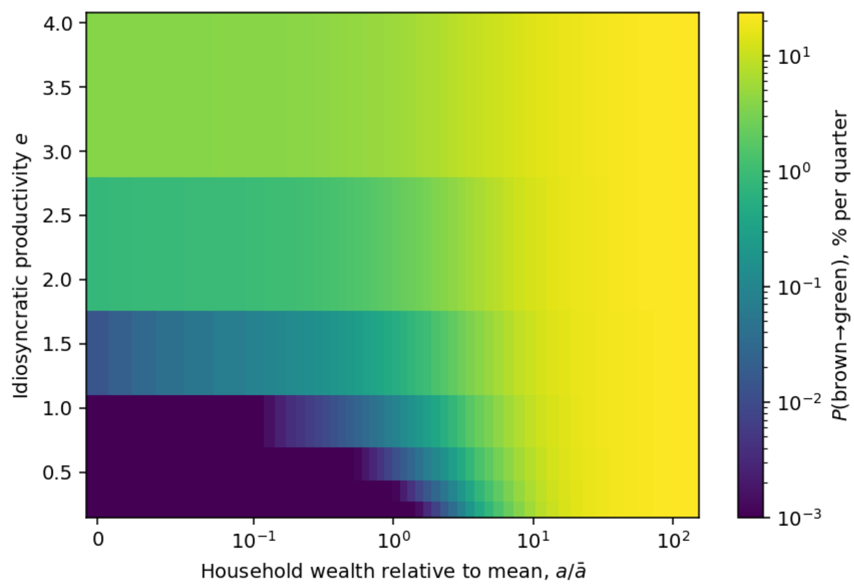

Figure 1: Probability of switching from a brown to a green durable, per quarter, for the median discount-factor type, as a function of household wealth relative to the mean $a/\bar{a}$ (horizontal, symmetric-log scale) and idiosyncratic productivity e (vertical), at the baseline steady state. Colour on a log scale. Plotting a single type's policy function rather than the mass-weighted pool removes a composition artifact: pooling the three discount-factor types, whose switching rates differ by orders of magnitude, makes the wealth gradient non-monotone even though each type's rule is monotone. The policy function is monotone in both wealth and productivity.

The wealth gradient is the marginal-switcher condition (11) seen household by household. Figure 2 prices one switch the way the condition does: the upfront cost ψ_g against the energy saving $(\bar{P}^E_B - \bar{P}^E_G)\,\bar{c}^G_E$ accumulated over the durable's life, discounted by interest and survival; the limit of the plotted sum is the discounted-saving side of (11). Because energy use rises with wealth in levels, so does the saving. At mean wealth the saving never reaches ψ_g: the typical household stays brown not because it discounts the future heavily but because its energy bill is too small for the operating saving ever to repay the cost. At about $25\,\bar{a}$ the breakeven arrives at the durable's expected life, the accounting margin; at $40\,\bar{a}$ it arrives within four years, and adoption is comfortably worthwhile. The figure is an accounting cut: it holds the green incumbent's demand fixed and abstracts from consumption smoothing and the taste shocks that make the actual choice probabilistic. But it is the reason the holding rate climbs from under three percent to seventy-seven across the wealth distribution (Figure 3), and the object the crisis then perturbs: a brown price shock scales the saving flow up, moving each household's crossing left.

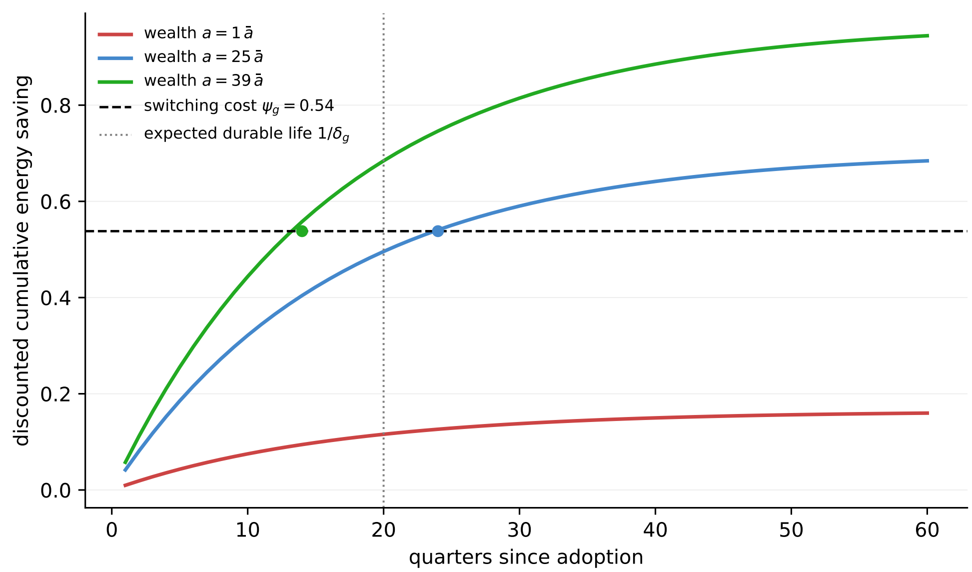

Figure 2: The private economics of one switch, at the steady state. For the median discount-factor type at median productivity and three wealth levels, cumulative energy saving from operating green, discounted by interest and survival, $S(T) = \sum_{t \le T} \left[(1-\delta_g)/(1+r)\right]^t (\bar{P}^E_B - \bar{P}^E_G)\, \bar{c}^G_E(e,a)$, against the switching cost ψ_g (dashed) and the expected durable life $1/\delta_g$ (dotted). Dots mark the breakeven quarter. No shock: the steady-state investment problem.

Two further steady-state objects anticipate the policy results (Figure 3). The switching *flow* has a stock counterpart: the green-holding rate rises from under 3% at the constraint to 77% at the top of the wealth grid, so the durable is held disproportionately by the wealthy. The marginal propensity to consume runs the other way, falling from about 0.5 at the constraint to near zero by $a \approx 3.5\,\bar{a}$, with brown incumbents and green holders close at each wealth level. These two gradients drive the incidence results of Section 6. A transfer to brown households reaches the high-MPC constrained tail and stabilises consumption effectively; an energy-indexed transfer does not, because the energy bill is flat as a share of spending (3.8% at the bottom of the wealth distribution against 3.7% at the top) and so rises with wealth in levels, sending the largest payments to the low-MPC wealthy. Homothetic energy demand makes energy-indexed targeting regressive; the flat share is the assumption behind that result, and the non-homothetic robustness check in Section 6 relaxes it.

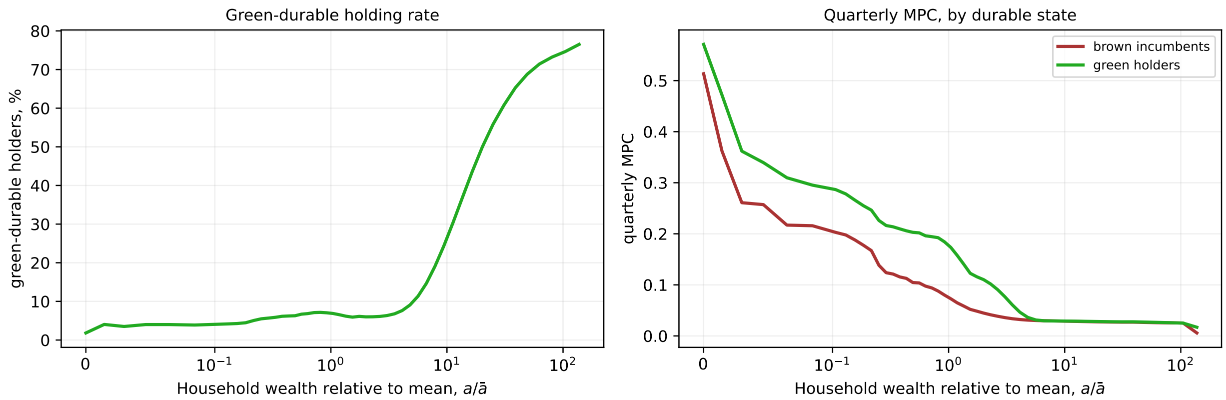

Figure 3: Steady-state objects, aggregated over discount-factor types. Left: green-durable holding rate against wealth (the stock behind the switching flow). Right: quarterly MPC against wealth, by durable state. Wealth on a symmetric-log scale.

5 The energy shocks and the adoption channel

5.1 The two shocks and the common-steady-state counterfactual

The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life, as in Auclert et al. (2024): energy supply is perfectly elastic ($\varepsilon^E = \infty$) and the world price is exogenous. The *supply* shock is a 10% cut in available energy for 6 quarters, after which the world price clears a fixed quantity endogenously, as in Bayer et al. (2026). The two are easiest to read side by side (Figure 4): under the price shock we move the world price directly and the quantity clears; under the supply shock we move the quantity, a 10% cut, and the world price clears it. Because energy demand is highly inelastic ($\eta_E = 0.10$), a 10% quantity cut requires a price spike of about 170% to clear, so the supply shock is in fact the larger *price* disturbance of the two on impact. Impulse responses are reported as 100× the level deviation from steady state. Variables whose steady state is one (prices, w, n, and the shares D^G, D^{sw}) read directly as percent or percentage-point deviations; C (steady state ≈ 0.997) is within rounding of a percent; y and gdp (steady state ≈ 0.985) are level deviations ×100, within about 1.5% of the exact percentage; energy quantities are a share of the energy steady state (≈ 0.040).

Several results below isolate the adoption channel as the difference between the full model and a "no-adoption" counterfactual, $\Delta_H \equiv \sum_{t<H}(y_t^{\text{adopt}} - y_t^{\text{no-adopt}})$. The counterfactual is constructed to share the full model's steady state *exactly*: households still hold the discrete durable-choice margin, and the steady-state green share, switching flow, and switching cost ($\bar{D}^G = 5\%$, D^{sw}_{ss}, ψ_g) are bit-identical to the adoption economy. What differs is the *transition*

only: the discrete-choice probabilities are held at their steady-state values throughout the impulse response, so the durable composition does not respond to the crisis while every other block responds normally. We call this the **common-steady-state**, or *frozen-choice*, counterfactual. The alternative, shutting the choice by making green durables infeasible, so the counterfactual's own steady state has $D_{ss}^{G} = 0$, would compare two economies that differ both in the crisis response *and* in steady-state composition; the frozen construction isolates the crisis response alone.

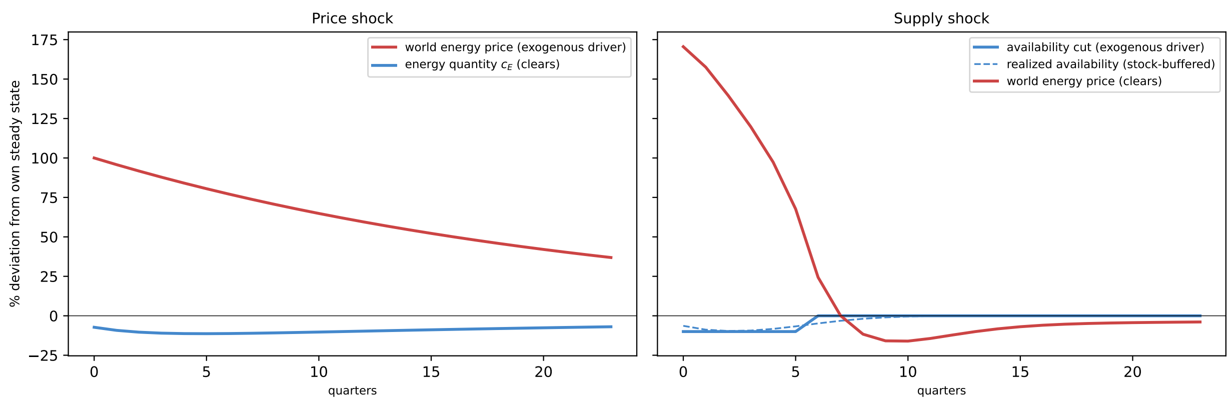

Figure 4: The two energy shocks. *Left*: the price shock moves the world energy price (exogenous) and the quantity clears. *Right*: the supply shock moves availability (−10% for 6 quarters, exogenous) and the world price clears the reduced quantity, spiking about 170% because demand is inelastic. In the right panel the bold blue line is the announced availability cut and the dashed blue line is realized availability: the two differ because world stock arbitrage (23) releases inventories against the cut, buffering part of it, and rebuilds them afterwards. Percent deviations from own steady state.

Figures 5 and 6 report the baseline responses to the two shocks, each with the adoption margin open (solid) against the frozen counterfactual (dashed).

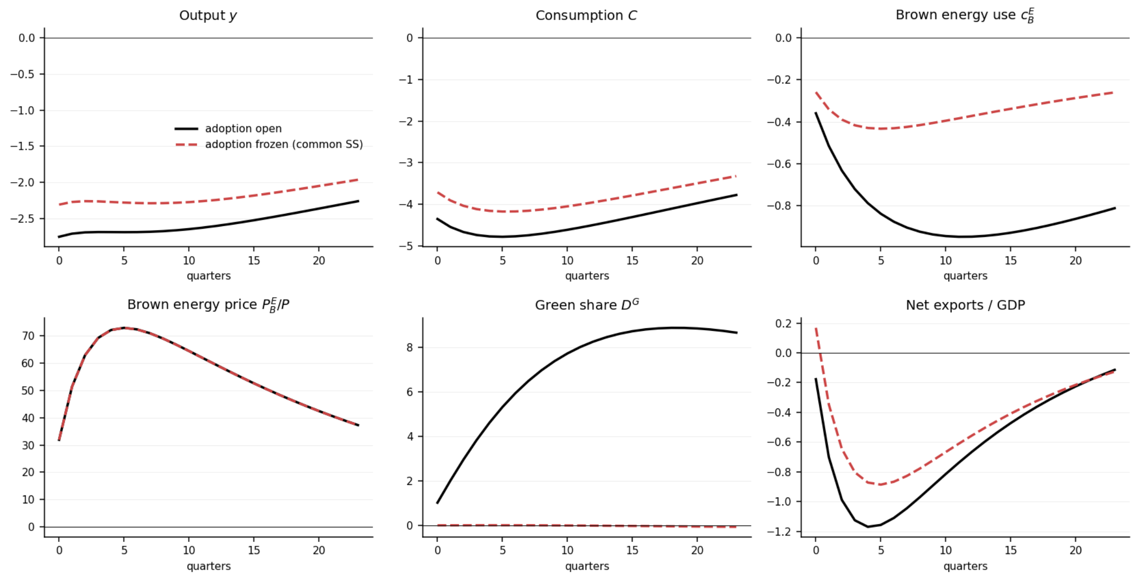

Figure 5: Brown energy *price* shock (elastic supply), adoption open (solid) vs. frozen at the common steady state (dashed), no policy.

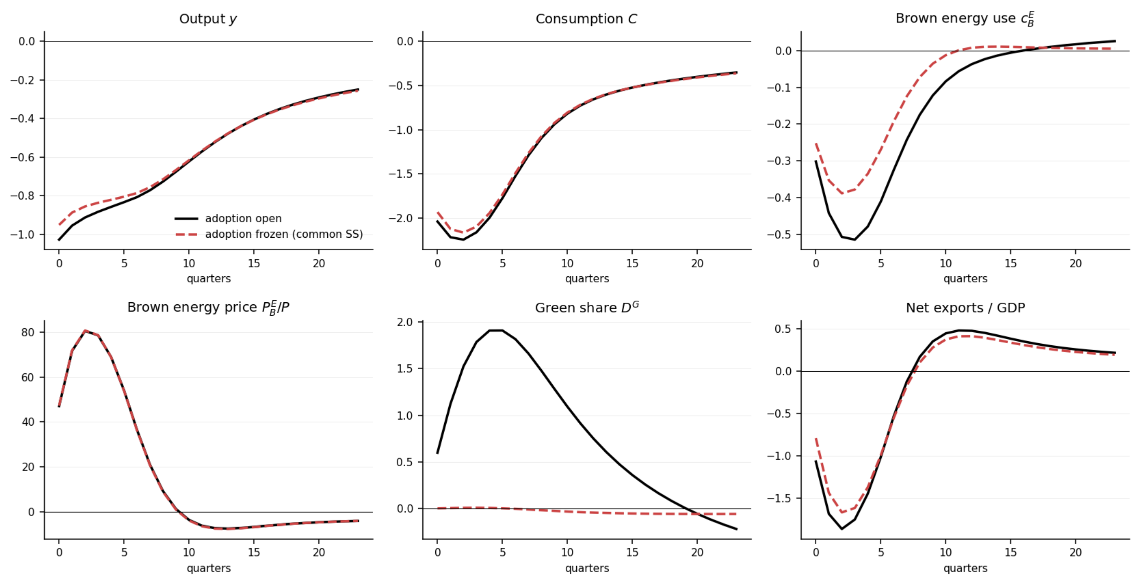

Figure 6: Brown energy *supply* shock (−10% for 6 quarters, fixed quantity), adoption open (solid) vs. frozen (dashed), no policy.

5.2 What the adoption margin adds

Under the import booking and the price shock, the adoption channel's contribution to cumulative output is $\Delta_{24} = -8.8$ (Table 4; Figure 7 plots the margin's contribution across variables): the margin is *contractionary*. It follows a race between two resource flows in the balance of payments (Section 3.4). Adoption imports a **switching cost** $\psi_g D_t^{sw}$, a resource outflow, *front-loaded* (paid once by each switcher) and scaling with the adoption *volume* D^{sw}, and, against it, a **flow energy saving** $(P_B^E - P_G^E)\, C_{E,t}^G$ booked as reduced imports, an inflow accruing *gradually* over the life of the green durable. Under the frozen counterfactual and the import booking the front-loaded cost dominates the windowed saving at every persistence. The magnitude is the pocket cost of the crisis-time switching wave at the calibrated $\psi_g = 0.54$ ($\approx 0.54 \times C_{ss}$), not a general-equilibrium artefact: the direct cost $\psi_g \times$ extra switchers cumulates to $+8.96$, matching the measured gap almost exactly. Under the fixed-quantity (supply) closure a third, expansionary term appears, adoption lowers fossil demand against a fixed supply, so the clearing price falls and relaxes scarcity for *all* households, which shifts Δ_{24} up to -0.35. The marginal switcher's steady-state indifference, $\psi_g \approx (\bar{P}_B^E - \bar{P}_G^E)\, \bar{C}_E^G/(r + \delta_g)$, ties the threshold to the cost ratio ϱ.

Booking caveat. The *sign* of Δ_{24}, not the policy ordering below, depends on how the green resource flows are booked. Under the alternative domestic-industry booking (green energy and installation as zero-profit domestic value added rather than imports) it flips to $+28.6$ under the price shock and $+4.7$ under the supply shock. We report the import booking throughout and collect the comparison in Appendix A. This is a booking convention, not a result about the transition, and it is why we phrase the paper's headline in terms of the *margin's response* to policy rather than its output sign.

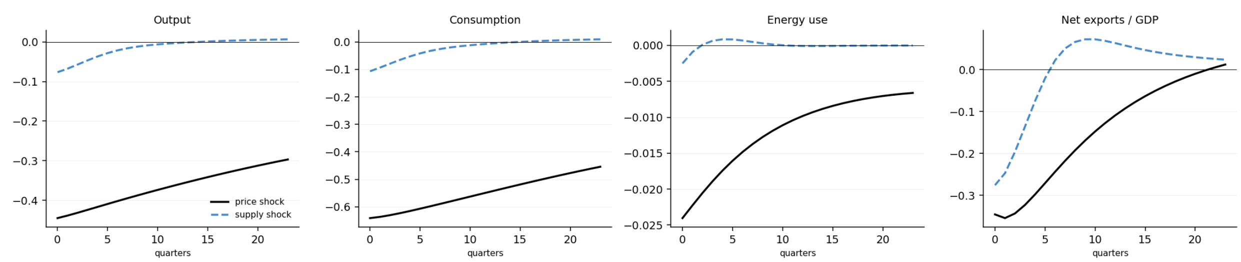

Figure 7: Contribution of the green adoption margin (full minus frozen), by shock. Output, consumption, energy use, net exports.

The race between the two resource flows is exact on the import side. Figure 8 splits the adoption channel's contribution to imports, full minus frozen, under the baseline import booking, into the front-loaded switching-cost outflow $\psi_g D_t^{sw}$ and everything else, the latter dominated by the gradual energy saving from the brown-to-green mix shift. The split is exact by construction. The switching cost dominates the saving at every horizon under both shocks: under the price shock it cumulates to $+8.96$ against a -5.63 energy-mix inflow, a net import response of $+3.33$ that accounts for the channel's contractionary output contribution

($\Delta_{24} = -8.8$). The outflow is largest exactly on impact, when household income is falling hardest, which is why booking it as an import makes the channel contractionary rather than its timing incidental.

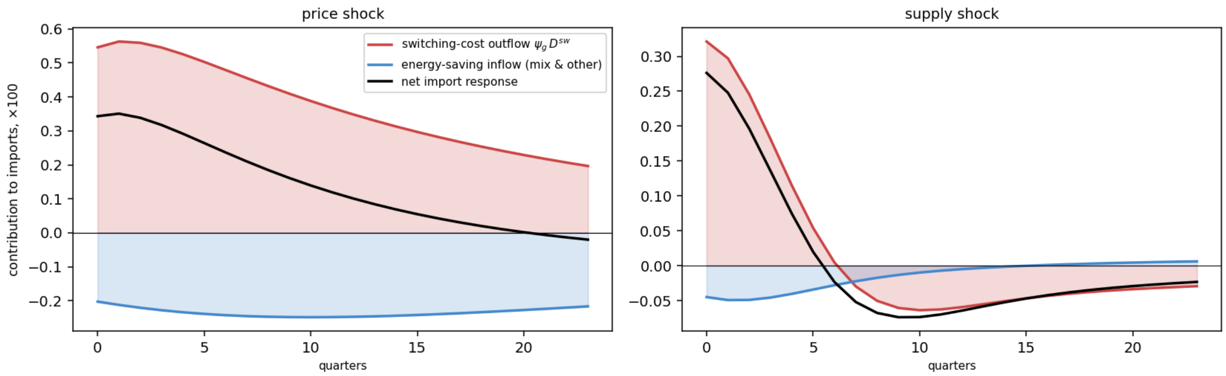

Figure 8: Adoption channel on the import side (full minus frozen): the front-loaded switching-cost outflow $\psi_g D^{sw}$ against the gradual energy-saving inflow, and their sum. Contributions in level deviations $\times 100$; import booking.

The adoption channel's output effect is a domestic-demand phenomenon. Decomposing output one level below the domestic/export split (Figure 9), the frozen-choice counterfactual separates domestic demand into what it would be with the margin shut, the AMRS real-income and substitution channels, and the adoption channel's increment. The adoption channel's cumulative contribution (-9.0) falls almost entirely on domestic home-good demand, not exports ($+0.2$): the switching-cost import outflow draws resources out of domestic absorption. Its magnitude nearly matches the export offset ($+8.7$) with the opposite sign, so aggregate output tracks the AMRS channel while the adoption drag and the competitiveness gain roughly cancel.

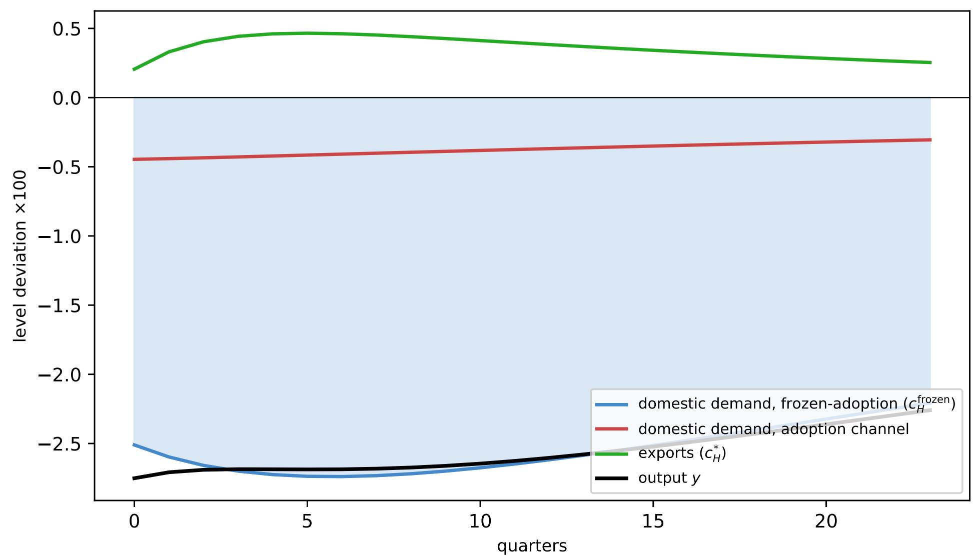

Figure 9: Output response, three exact channels: exports c_H^*, domestic demand with the adoption margin frozen (AMRS), and the adoption channel's increment to domestic demand. The three sum to y. Level deviations $\times 100$, price shock, no policy.

The same response, read from the household budget rather than from output, is almost entirely a general-equilibrium phenomenon. Decomposing consumption through the household consumption Jacobians, $dC = \sum_x J^{C,x}\, dx$ over the household's inputs, exact to first order, separates the direct effect of the shock, the energy bill in the budget, from the indirect effects that run through factor income and the real rate (Figure 10, computed in the CPI numeraire where these inputs are unambiguous; the total is numeraire-invariant). The direct energy-price channel is negligible (-0.2 cumulative): energy is a small budget share, so a higher energy price does little to consumption on its own. The collapse is the indirect real-income channel (-101.8 of a -105.6 total), with the real rate adding -3.6. The energy shock reaches consumption almost entirely by lowering factor income in general equilibrium, not by raising the household's energy bill, the mechanism Auclert et al. (2024) and the broader HANK literature identify, here in an open economy with an adoption margin.

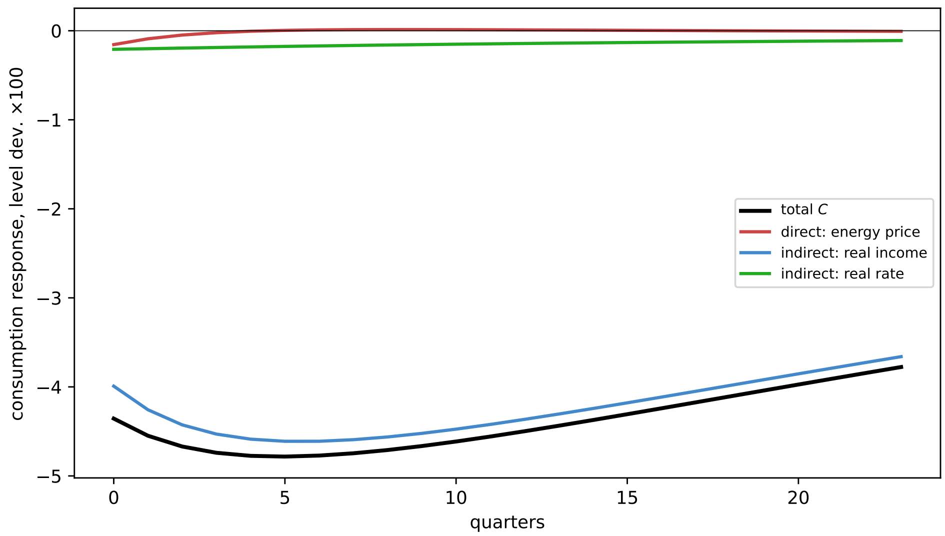

Figure 10: Consumption response to the brown-price shock, decomposed into the channels through which the shock reaches the household budget: the direct energy-price effect and the indirect effects operating through real income and the real rate. Household consumption Jacobians, CPI numeraire; the channels sum to the total. Level deviations ×100.

Who the shock recruits. The crisis draws in households who would not have switched at the pre-crisis steady state. Figure 11 characterises them with a comparative static: the switching probability of the median type at a steady state with a permanently 10% higher brown price, minus the baseline field. The induced switching is positive throughout and rises with productivity; at high productivity the recruitment reaches down to more moderate wealth, pulling in households who sat just below the switching threshold. Weighted by the steady-state distribution, the induced flow concentrates at moderate productivity and low-to-moderate wealth, where the brown mass sits. The construction is a comparative-static proxy: it uses a permanent price change rather than the transitory crisis path, so it locates and signs the induced margin rather than timing it; the exact transition object reads the disaggregated choice probabilities along the impulse response.

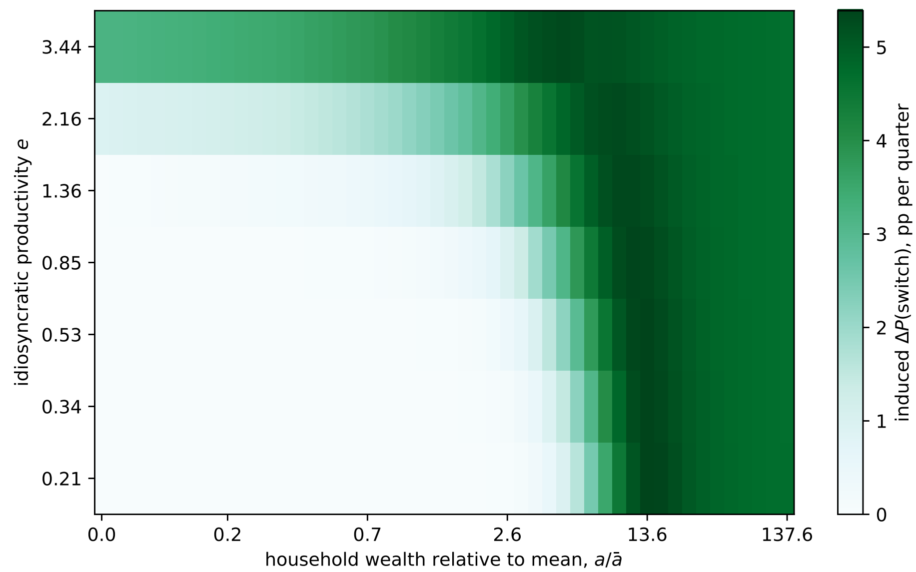

Figure 11: Induced adopters: the extra quarterly switching probability a permanently 10% higher brown price induces, median discount-factor type, over the wealth–productivity grid (comparative static). Contours mark steady-state brown mass.

Shock	Policy	$y(0)\%$	$\sum y$	$\pi(0)$ ann.	peak D^G pp	fiscal cost	adoption $\to \sum y$
price	none	-2.75	-61.5	$+7.32$	**8.87**	0.00	-8.79
	subsidy	-1.66	-53.0	-3.37	**-0.04**	53.86	$+0.69$
	transfer	-0.63	-22.3	$+9.09$	**9.15**	49.27	-8.93
supply	none	-1.03	-13.9	$+8.86$	**1.91**	0.00	-0.35
	subsidy	-1.72	-45.6	-1.41	**-0.03**	-18.45	$+0.09$
	transfer	$+0.56$	-8.5	$+12.00$	**2.59**	17.74	$+0.22$

Table 4: Cumulative sums over 24 quarters. The last column is the contribution of the adoption margin: the difference between the full model and a common-steady-state counterfactual in which the adoption choice does not respond to the shock (Section 3).

6 Shielding consumers: cap versus transfer

Both instruments in the recent literature stabilise the crisis, and Table 4 shows they do so comparably. They part ways on the margin this paper adds.

6.1 The cap eliminates the adoption margin

Under the cap the peak green-share response falls from 8.9 pp to essentially zero (−0.04 pp) under *both* shocks: the margin is not attenuated, it is switched off. The mechanism is direct: the cap holds P_B^E at its pre-crisis level, and P_B^E is precisely what determines the return to adoption. The full fiscal responses under both shocks are in Figures 12 and 13 (no policy, price cap, Slutsky transfer).

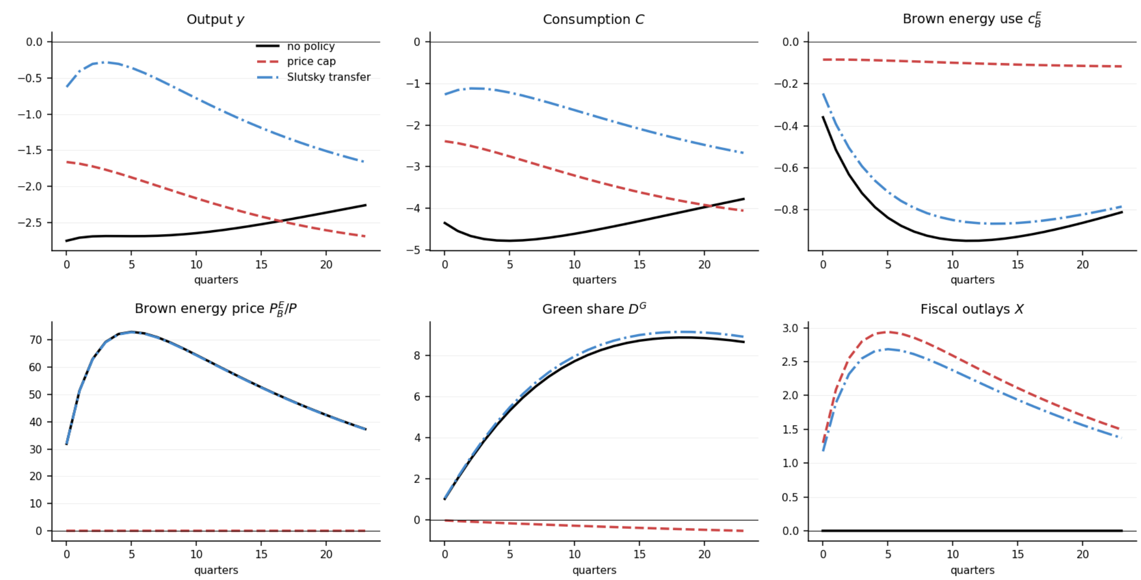

Figure 12: Fiscal response to the *price* shock, adoption open: no policy (solid), price cap (dashed), Slutsky transfer (dash-dotted).

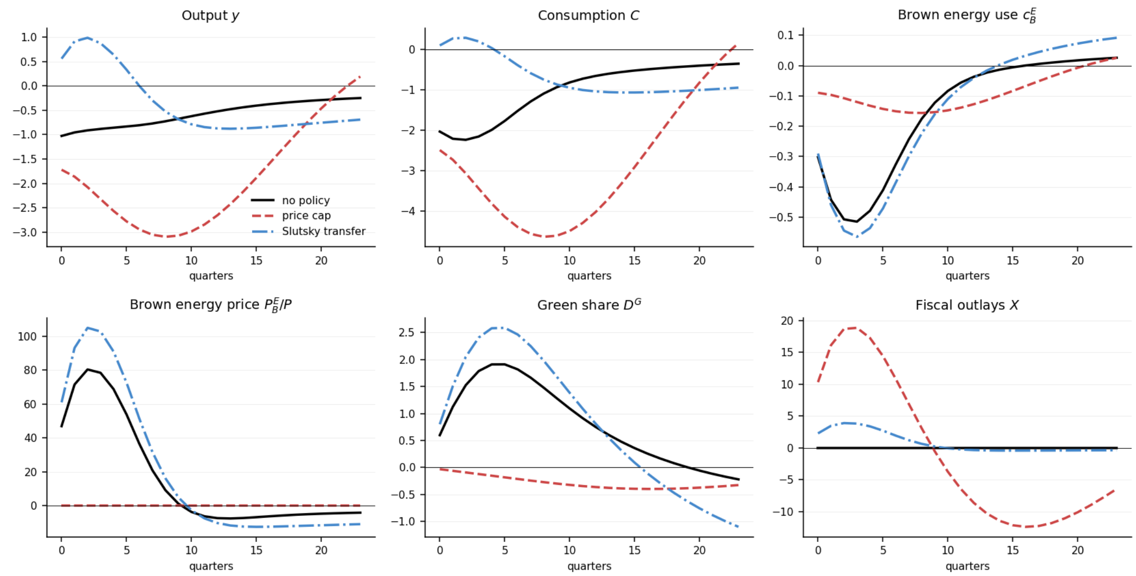

Figure 13: Fiscal response to the *supply* shock, adoption open: no policy (solid), price cap (dashed), Slutsky transfer (dash-dotted).

6.2 The transfer preserves it

The Slutsky transfer leaves the green-share response essentially unchanged (9.1 pp against 8.9 with no policy) while delivering more output stabilisation than the cap at similar fiscal cost ($\sum y = -22.3$ vs. -53.0; gross fiscal 49.3 vs. 53.9). This is a direct consequence of the transfer's lump-sum indexing (Section 3.4): because it does not condition on the household's current durable, switching forfeits nothing and the relative-price incentive to adopt is left intact. Subsidies destroy the transition margin; transfers do not.

6.3 A second casualty: the expenditure-switching channel

The relative price the cap freezes drives more than adoption. Decomposing the output response through the open-economy identity $y = c_H + c_H^*$, home-good output equals domestic demand for the home good plus exports, makes the point (Figure 14). Under no policy the energy shock lifts exports: the relative price of the home good falls and foreign demand rises, a partial offset to the domestic-demand collapse (cumulatively $+7.4$ against a domestic -72.1). The cap switches that offset off almost entirely ($+0.7$): holding the retail energy price fixed suppresses the relative-price movement expenditure switching runs on, exactly as it suppresses the one adoption runs on. The transfer leaves the relative price free and preserves the export channel nearly intact ($+6.1$) while supporting domestic demand (-31.6). The cap and the transfer stabilise comparable amounts of output, but the cap does so by suppressing an adjustment margin and the transfer by supporting income: the same distinction the paper draws on the adoption share, read now through the trade balance.

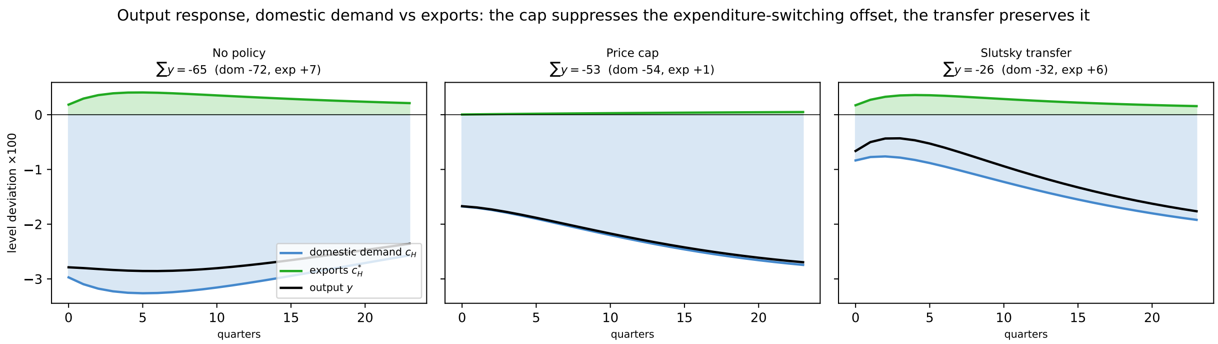

Figure 14: Output response decomposed as $y = c_H + c_H^*$ (domestic demand for the home good plus exports), by policy, under the brown-price shock. The export offset (green) is positive under no policy and the transfer and switched off under the cap. Level deviations $\times 100$.

6.4 The cap is a dial, not a switch

Table 4 uses $\tau^E = 1$, a complete price freeze; Table 5 sweeps $\tau^E \in [0, 1]$. Under the *price* shock the adoption loss is almost exactly linear in τ^E. Under the *supply* shock it is convex: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial shield is therefore much less damaging to the transition when supply is inelastic, but, as the next result shows, much more damaging to everything else.

Shock	Policy	peak D^G pp	$\sum y$	$\pi(0)$	$\sum c_E$	fiscal	CEV %
price	cap $\tau^E = 0.00$	8.871	-61.46	7.32	-8.94	0.00	-1.8710
	cap $\tau^E = 0.25$	6.547	-59.36	4.68	-7.50	13.40	-1.6917
	cap $\tau^E = 0.50$	4.218	-57.25	2.01	-6.06	26.85	-1.5112
	cap $\tau^E = 0.75$	1.894	-55.15	-0.67	-4.62	40.34	-1.3296
	cap $\tau^E = 1.00$	-0.036	-53.04	-3.37	-3.17	53.86	-1.1469
	transfer	9.146	-22.27	9.09	-6.43	49.27	-0.8162
supply	cap $\tau^E = 0.00$	1.911	-13.94	8.86	-2.40	0.00	-0.6280
	cap $\tau^E = 0.25$	1.754	-16.89	7.99	-2.40	4.61	-0.6694
	cap $\tau^E = 0.50$	1.503	-21.48	6.71	-2.41	11.01	-0.7355
	cap $\tau^E = 0.75$	1.031	-29.64	4.52	-2.43	18.36	-0.8545
	cap $\tau^E = 1.00$	-0.033	-45.55	-1.41	-2.79	-18.45	-1.0843
	transfer	2.588	-8.49	12.00	-2.40	17.74	-0.2881

Table 5: Dose-response in the price-cap intensity τ^E, against the Slutsky transfer benchmark (full compensation, $\iota = 1$).

6.5 Welfare, and the supply-elasticity sign flip

Under the supply shock the cap yields cumulative output of -45.6 against -13.9 with no policy: with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief. This survives the addition of an adoption margin.

The welfare measure. We rank scenarios by the discounted utility of a household drawn at random before the shock. There are three permanent discount-factor types $i \in \{0, 1, 2\}$ of equal mass; type-i welfare in a scenario is

$$\mathcal{W}_i = \sum_{t \geq 0} \beta_i^t \Big(\mathbb{E}_0[u(c_{it})] - v(n_t) \Big), \tag{33}$$

the expectation over the date-0 (pre-shock) cross-section of type i; the distribution then evolves endogenously along the transition, so (33) is the expected discounted utility of a household placed behind a date-0 veil of ignorance. This is the object a welfarist planner evaluates; Dávila and Schaab (2025) give the general treatment of welfare assessments with heterogeneous individuals. Auclert et al. (2024), the model we extend, gauge the distributional effects of energy policy through consumption inequality; we complement that with a consumption-equivalent welfare ranking. We report $\mathcal{W}_i$ type by type and, as a population summary, the equal-weighted mean across the three types. Felicity u is (2), evaluated on the bundle the household consumes, so $\mathbb{E}_0[u(c_{it})]$ is the exact cross-sectional average, not felicity at the average bundle.

Labour is not a household choice. The union sets the wage (14) and hours n_t are common across households: a household of productivity e earns $e\,(1-\tau^L)w\,n_t$ and works the same hours as everyone else. Preferences are separable and n_t is taken as given, so the labour-disutility term $v(n) = v_\varphi\, n^{1+\varphi}/(1+\varphi)$ does not enter the household's consumption–saving problem (5). It is nonetheless a real utility cost of the hours worked, common to every type, so the welfare criterion must include it: we subtract $v(n_t)$ in (33). Omitting it would score a policy that raises employment as a free lunch (higher consumption with no offsetting toil), which it is not.

From the transition to the CEV. Along a scenario's transition the felicity deviations are $\widehat{u}_{it} = u_{it} - u_i^{\mathrm{ss}}$, where $u_{it} = \mathbb{E}_t[u(c_{it})]$ is average type-i felicity at date t and u_i^{ss} its reference-steady-state level; the labour term deviates by $\Delta v_t = v'(n^{\mathrm{ss}})\,\widehat{n}_t$, with $v'(n^{\mathrm{ss}}) = v_\varphi\,(n^{\mathrm{ss}})^\varphi$. The scenario's welfare relative to its own steady state is

$$\Delta\mathcal{W}_i = \sum_{t\geq 0} \beta_i^t \left(\widehat{u}_{it} - \Delta v_t\right). \tag{34}$$

We express this in consumption units through the consumption-equivalent variation χ_i (Lucas, 1987): the permanent proportional consumption supplement that, granted to every type-i household in the reference steady state, reproduces $\Delta\mathcal{W}_i$. A permanent supplement scales felicity by $(1+\chi_i)^{1-\sigma}$ and so raises welfare by $[(1+\chi_i)^{1-\sigma}-1]\,u_i^{\mathrm{ss}}/(1-\beta_i)$; equating to $\Delta\mathcal{W}_i$ and inverting,

$$\chi_i = \begin{cases} \exp\Big[(1-\beta_i)\,\Delta\mathcal{W}_i\Big] - 1, & \sigma = 1, \\ \Big[\,1 + (1-\beta_i)\,\Delta\mathcal{W}_i/u_i^{\mathrm{ss}}\Big]^{1/(1-\sigma)} - 1, & \sigma \neq 1. \end{cases} \tag{35}$$

We report χ_i by type and averaged, in percent of permanent consumption; $\chi_i < 0$ is a loss, the share of consumption a household would forgo in perpetuity to avoid the scenario. Because (34)–(35) use real felicity, the CEV does not depend on the unit of account.

A common reference across steady states. Scenarios that begin from different steady states cannot be compared through (34) alone: the *ex-post* crisis instruments start at the pre-crisis baseline, whereas the *ex-ante* carbon tax has already moved the economy to a greener, permanently distorted steady state before the crisis. We fix the baseline steady state as the common reference and split each scenario's welfare into a standing and a transition term,

$$\Delta\mathcal{W}_i = \underbrace{\frac{u_i^{\text{pre}} - u_i^{\text{base}} - \Delta v^{\text{lvl}}}{1-\beta_i}}_{\text{standing gap vs. baseline}} + \underbrace{\sum_{t\geq 0} \beta_i^t \Big(\widehat{u}_{it} - \Delta v_t\Big)}_{\text{crisis transition}}, \tag{36}$$

where u_i^{pre} is felicity at the scenario's own pre-crisis steady state, u_i^{base} at the baseline, and $\Delta v^{\text{lvl}} = v'(n^{\text{base}})\,(n^{\text{pre}} - n^{\text{base}})$ the standing labour-disutility difference. The standing gap is zero for scenarios that start at the baseline and negative for the ex-ante tax, which gives up steady-state consumption to raise adoption; carrying it into $\Delta\mathcal{W}_i$ places ex-ante and ex-post policies on a single welfare axis. We feed (36) through (35) exactly as for any other scenario.

First-order accuracy. The deviations in (34) are first order in the shock, so χ_i is the leading term of the welfare expansion around the (distorted) steady state along a deterministic transition. This is not the stochastic setting in which a first-order evaluation can reverse welfare rankings (Kim and Kim, 2003; Benigno and Woodford, 2005): the transition is deterministic and the first-order term is nonzero and dominant. Because the shock is large and the adoption margin is convex, we bound the neglected second-order terms by recomputing every χ_i on the nonlinear transition (Appendix B.2); the levels move by at most 10–15% and no ranking changes.

Table 6 turns to welfare, measured as the total consumption-equivalent variation (CEV) relative to the common no-tax steady state. On private welfare the Slutsky transfer dominates the cap under *both* shocks (price: -0.82% vs. -1.15%; supply: -0.29% vs. -1.08%); the case for the transfer over the cap does not rest on the transition. But the *direction* of the cap's welfare effect turns on the supply elasticity: tightening the cap raises welfare under the price shock and lowers it under the supply shock (Figure 15).

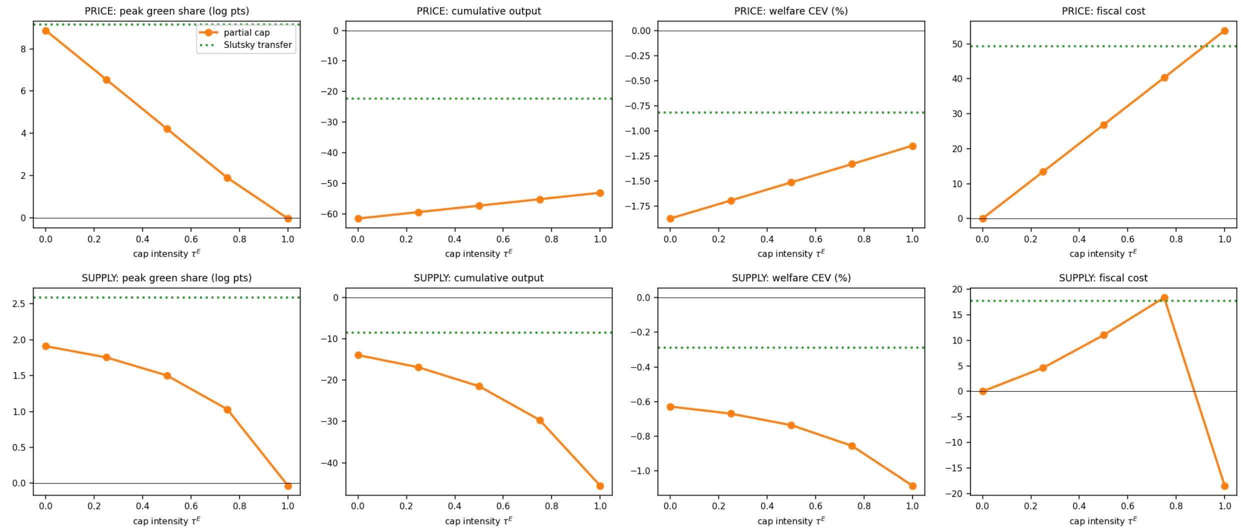

Figure 15: Dose-response in τ^E (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

Scenario	$y(0)\%$	$\sum y\%$	peak D_G	gross fisc.	CEV%
Laissez-faire	-2.75	-61.5	**8.87**	0.0	-1.871
Ex-post cap	-1.66	-53.0	**-0.04**	53.9	-1.147
Slutsky transfer	-0.63	-22.3	**9.15**	49.3	-0.816
Flat transfer	-0.31	-18.4	**9.10**	49.3	-0.317
Green subsidy	-3.19	-70.1	**18.02**	2.1	-2.032
Ex-ante ETS ($\tau_b = 0.1$)	-2.85	-62.2	**14.98**	0.1	-1.852

Table 6: Price-shock policy responses (import booking). $y(0)$ impact output, $\sum y$ cumulative output over $H = 24$, peak green share, gross fiscal disbursement, and total CEV vs the no-ETS baseline SS.

Locating the Langot–Bayer contrast. Langot et al. (2026), assuming elastic supply, find the French shield effective; Bayer et al. (2026), assuming fixed union-wide supply, find subsidies destructive. Our two closures reproduce the *sign* each obtains, which locates one source of the contrast in a single parameter. The narrow point is that, once the supply elasticity is fixed, the direction of the cap's welfare effect is determined, so part of the empirical disagreement reduces to measuring that elasticity at the relevant horizon.

Private welfare and the transition are distinct. The CEV above is a *private* household-welfare measure: it values consumption and leisure along the transition and prices no climate or transition externality. Two results are therefore kept separate. On private welfare alone the transfer dominates the cap under both shocks; the case for the transfer does not rest on the transition. Separately, the transfer preserves the green adoption margin the cap eliminates (Section 6.1): a positive statement about the transition, not a welfare ranking. We do not aggregate the two into one social-welfare number: that would require a social value of adoption (a carbon price times avoided brown-energy use) we do not model. A positive such value only reinforces the private ranking; it cannot reverse it, since the transfer already dominates.

6.6 Distributional incidence and regressive targeting

Without policy the impatient (high-MPC) households lose far more than the patient (Table 7). This makes the incidence of the transfer a first-order design question. The Slutsky transfer indexes the lump sum to each household's pre-crisis brown-energy use. Replacing it with an *untargeted* flat transfer of the *same aggregate envelope*, the identical outlay (49.3 vs. 49.3 over $H = 24$), paid flat rather than indexed, almost halves the welfare loss, from a CEV of -0.82% to -0.32% (Table 6), and even improves output ($\sum y = -18.4$ vs. -22.3), while leaving adoption intact (9.1 pp). The gain is entirely distributional.

The mechanism is that energy-indexed targeting is regressive *in this model*. With homothetic CES energy demand the energy share is constant across the wealth distribution, so brown-energy use rises with total consumption, hence with permanent income and patience. Indexing the transfer to pre-crisis energy use therefore hands the largest cheques to the low-MPC, wealthy households, exactly where a euro buys the least insurance, while a flat transfer reaches the high-MPC households the crisis hits hardest. This speaks to the 2022–23 European energy "brakes," which were indexed to historical consumption: within the model the indexing is the source of the inefficiency, not a refinement of it.

We flag the caveat plainly. The result rests on the homothetic energy nest: if energy were a necessity with a declining Engel curve, the empirically standard case, low-income households would spend a *larger* share and the sign could reverse. The clean statement is conditional: *under constant energy shares, energy-indexed targeting is dominated by an equal-value flat transfer.*

Scenario (price shock)	CEV mean	impatient	middle	patient
no policy	-1.87	-2.37	-2.05	-1.20
cap $\tau^E = 0.5$	-1.51	-1.71	-1.65	-1.17
cap $\tau^E = 1$	-1.15	-1.05	-1.25	-1.14
transfer	-0.82	-0.68	-0.89	-0.88

Table 7: CEV by discount-factor type (%). Impatient types hold little wealth and have high MPCs. Private household welfare only.

7 Ex ante or ex post? The adoption absorber

The instruments in Section 6 are all *ex post*: deployed once the shock has landed. Because the adoption margin responds to the same relative price the crisis moves, a natural alternative is to act *ex ante*, to lean on the margin in the steady state with a permanent carbon tax τ_b on brown energy, leaving the economy greener and, one might hope, less exposed. This section prices that hope. The answer is that ex-ante greening is a poor substitute for ex-post insurance, and the reason is intrinsic to the channel this paper adds: the adoption margin is an endogenous shock absorber, and pre-greening spends it.

7.1 A welfare-optimal permanent carbon tax

Holding ψ_g fixed at its no-tax value and letting D^G_{ss} float, a carbon tax raises the effective brown/green price ratio and lifts steady-state adoption: D^G_{ss} moves from 5% at $\tau_b = 0$ to 8.9% at $\tau_b = 0.10$ and 20.1% at $\tau_b = 0.30$ (Figure 16, right). Steady-state welfare is hump-shaped in τ_b (Figure 16, left), rising to a peak of +0.043% at $\tau_b^* \approx 10\%$ and turning negative beyond $\tau_b \approx 25\%$. The model contains *no* climate externality, so this interior optimum is not Pigouvian. It is a second-best correction of the adoption friction: green energy is structurally cheaper ($\bar{P}^E_G = 0.8 < \bar{P}^E_B = 1$) but the logit taste scale σ_ε and the switching cost ψ_g leave green under-adopted, so a modest tax raises welfare until the tax distortion dominates. We treat this as a diagnostic of the channel, not a policy recommendation: it would be swamped by any explicitly modelled externality or abatement cost. The carbon revenue is rebated lump-sum here; Appendix A.2 studies the budget-neutral alternative that earmarks it to a green switching subsidy, which greens the steady state *more* yet leaves a residual brown pool *less* responsive to the crisis: the same absorber logic as Section 7.4, across instruments.

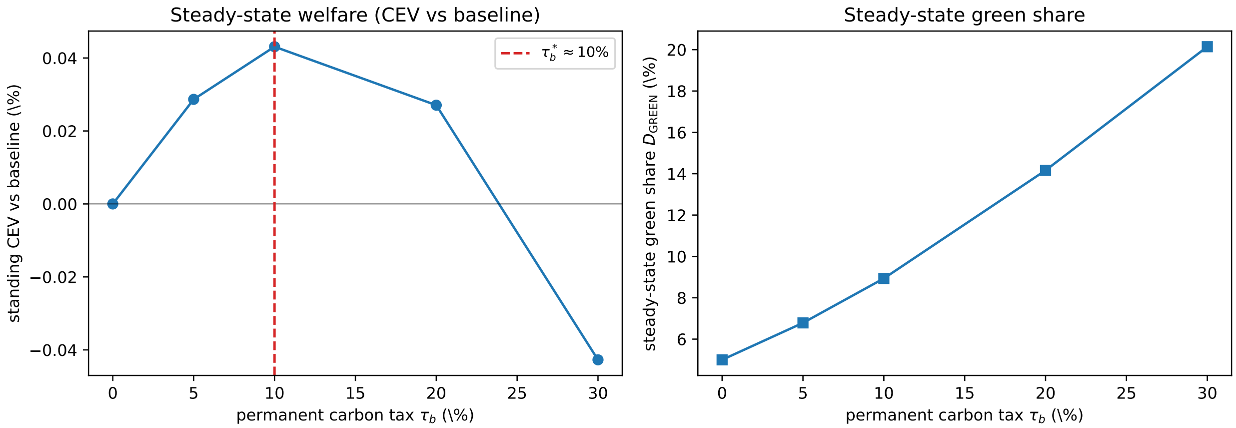

Figure 16: The welfare-optimal permanent carbon tax. Left: standing CEV of the τ_b steady state relative to the no-tax baseline (no crisis), hump-shaped with an interior optimum near $\tau_b^* \approx 10\%$. Right: the induced steady-state green share D_{ss}^G. ψ_g is held fixed; the carbon revenue is rebated lump-sum.

7.2 Ex-ante greening barely insures

We now hit two economies with the *same* price shock: the baseline ($D_{ss}^G = 5\%$) and a "prepared" economy carrying the welfare-optimal $\tau_b = 0.10$ ($D_{ss}^G = 8.9\%$). The prepared economy is essentially no better insured: its crisis CEV is -1.85% against -1.87% for laissez-faire, a difference of two basis points, and both are far worse than either ex-post instrument (cap -1.15%, transfer -0.82%; Table 9). The greening the tax buys is real, D_{ss}^G is 78% higher, but too small a share of the economy to move aggregate exposure, and the tax's crisis-time value is a rounding error next to a transfer that puts cash in high-MPC hands exactly when it binds. Consistent with Section 6.1, the combination "ETS + cap" tracks the cap alone: once the price is frozen the adoption margin is shut and the greener starting point is irrelevant.

The two economies are worth seeing side by side. Figure 17 overlays their responses to the same brown-price shock. The aggregates, output, consumption, inflation, net exports, are nearly indistinguishable; the two differ visibly only in brown energy use, the green share D^G, the switching flow, and the brown price. The greener starting point changes the *composition* of the response, not its aggregate size.

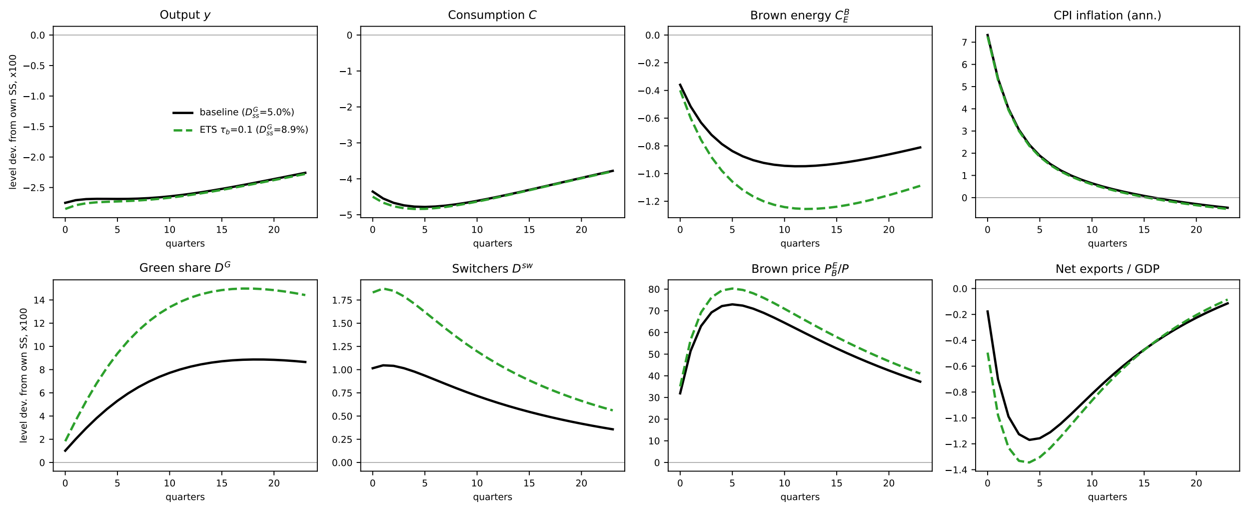

Figure 17: The same brown-price shock hitting the baseline ($D_{ss}^G = 5\%$) and the prepared ETS ($\tau_b = 0.10$, $D_{ss}^G = 8.9\%$) steady states, in level deviations from each economy's own steady state. Aggregates coincide; brown energy, the green share, switching and the brown price differ.

Figure 18 and Table 8 decompose the consumption response by the household's current technology. Switchers consume roughly twice the average brown household in the steady state, so a naive current-group split is contaminated by selection; we use the common-steady-state counterfactual to hold group populations fixed, read the clean per-capita response of each group off the frozen path, and report the adoption/migration term (full minus frozen) separately. Same aggregate loss, different cross-section: under the ETS both brown and green incumbents lose *less* (per-capita -92% against -97%, and -28% against -37% of their own steady state over $H = 24$), and that relief is exactly offset by a heavier adoption term (-22.8 against -13.2, cumulative $\times 100$), more households paying ψ_g to switch during the crisis. Green incumbents, fully insulated from the brown price, still lose about 1.5% per quarter, through the general-equilibrium income channel rather than energy.

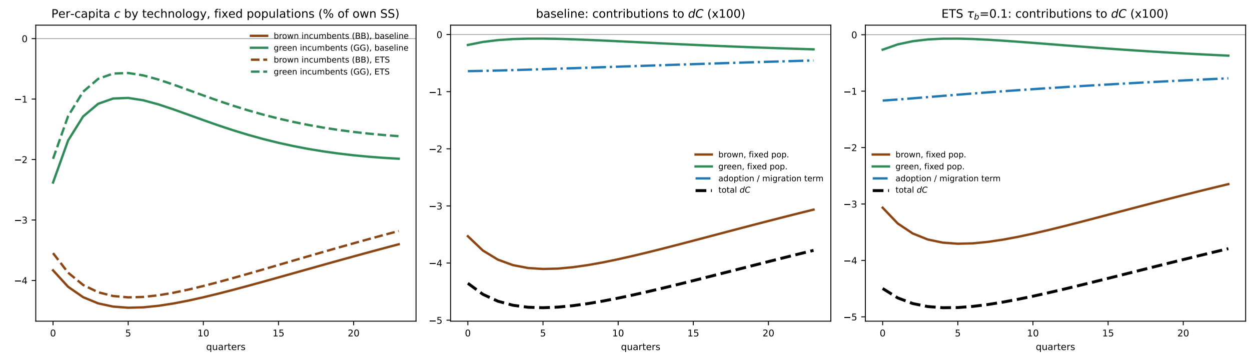

Figure 18: Consumption by technology group under the brown-price shock, baseline and ETS. Left: per-capita consumption of brown and green incumbents at fixed populations (% of own steady state). Centre and right: contributions to dC (fixed-population brown, fixed-population green, and the adoption/migration term).

Shock	Economy	D^G_{ss}%	$\sum dC$	brown (fixed)	green (fixed)	adoption term	$\bar{c}_B$%	$\bar{c}_G$%	peak D^G
price	baseline	5.0	-105.6	-88.7	-3.7	-13.2	-96.9	-37.4	**8.87**
price	ETS $\tau_b = 0.1$	8.9	-106.5	-78.7	-4.9	-22.8	-92.0	-27.8	**14.98**
supply	baseline	5.0	-24.5	-22.0	-1.9	-0.5	-24.8	-9.1	**1.91**
supply	ETS $\tau_b = 0.1$	8.9	-24.6	-20.8	-2.8	-1.1	-25.4	-6.5	**3.86**

Table 8: Consumption response by technology group over $H = 24$ (import booking, no policy). Cumulative level deviations $\times 100$: dC = brown (fixed population) + green (fixed population) + adoption term, where the fixed-population terms come from the common-steady-state frozen-choice counterfactual and the adoption term is full minus frozen. $\bar{c}_j$%: cumulative per-capita consumption response of brown / green incumbents in % of their own steady state, fixed populations.

Scenario	CEV%	impat.	middle	patient	peak D_G	gross fisc.
Laissez-faire	-1.871	-2.370	-2.048	-1.196	**8.87**	0.00
Ex-post cap	-1.147	-1.049	-1.253	-1.138	**-0.04**	53.86
Ex-post transfer	-0.816	-0.682	-0.886	-0.880	**9.15**	49.27
Ex-ante ETS	-1.852	-2.294	-2.016	-1.246	**14.98**	8.22
Ex-ante ETS + cap	-1.087	-0.942	-1.180	-1.137	**-0.06**	61.65

Table 9: Ex-ante greening vs ex-post capping under a brown-energy price shock (import booking, $\tau_b = 0.1$). CEV is the total consumption-equivalent variation relative to the no-ETS steady state, by discount-factor type; gross fiscal is the cap outlay (ex-post) or the standing carbon revenue over $H = 24$ (ex-ante).

7.3 The green subsidy at the trough

The green/adoption subsidy is worth studying as a crisis instrument in its own right (Figure 19): the government pays a fraction of the switching cost during the acute crisis. It roughly doubles the adoption response (peak D^G of 18.0 vs. 8.9 pp; switchers 2.3 vs. 1.0 pp) at trivial fiscal cost (2.0 against the cap's 53.9), yet its CEV is the worst in Table 6, -2.03%, below laissez-faire, and it deepens the output loss ($\sum y = -70.1$). Paying households to incur the real switching cost $\psi_g D^{sw}$, booked as an import (Section 3.4),in the middle of a consumption crisis destroys welfare even as it greens the economy. The adoption margin is valuable, but not as something to force at the trough.

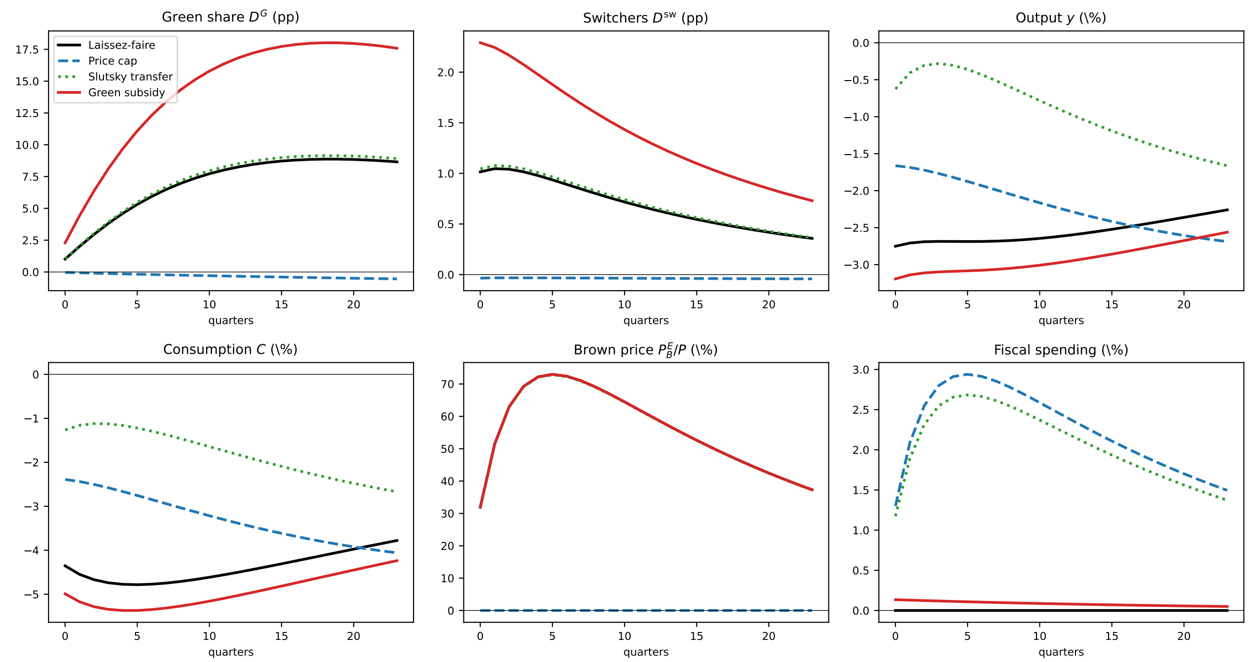

Figure 19: The green/adoption subsidy at the trough, against laissez-faire, the price cap and the Slutsky transfer (brown-price shock). It doubles the adoption response (D^G, D^{sw}) at trivial fiscal cost, yet forcing the real switching cost in the middle of the consumption crisis leaves output and private welfare below laissez-faire.

7.4 Persistence exhausts the absorber

The obvious defence of the ex-ante case is persistence: a longer-lived shock should let the greener economy's lower exposure compound over more periods. Figure 20 and Table 10 test it by sweeping the shock half-life and re-solving only the impulse response (the steady state does not move with persistence), decomposing the ex-ante number into its constant standing part and a *crisis-only greening dividend*, defined as the prepared economy's crisis CEV minus the standing benefit minus laissez-faire.

The defence fails, and instructively, though the failure is one of welfare, not of switching volume. The crisis-only greening dividend does not grow with persistence; if anything it falls with the half-life of the shock. Its level is at most a few basis points at every horizon, inside the first-order approximation error of the CEV (Appendix B.2), so we do not sign it: it is zero to first order. The ex-post cap is different in kind, its crisis protection rises monotonically, from +0.24% to +1.36% as a longer freeze shields more of the transition, an order of magnitude larger than the approximation error and robust nonlinearly.

The reason is the adoption margin itself. A persistent brown-price shock makes switching attractive for many quarters, so the *browner* laissez-faire economy adopts heavily *during* the crisis and reaps that windfall over the whole episode, the margin is an endogenous shock absorber. The prepared economy has already spent much of that absorber in the steady state. It is not that it adopts less in the crisis, with a standing tax it in fact adopts *more* (peak D^G of 15.0 against 8.9 pp; Section 7.2), but that the marginal crisis switch is worth less to it: the households the tax has already moved to green are inframarginal to the shock, and the extra crisis adoption pays the real cost ψ_g at the trough (Section 7.3). The welfare-relevant absorber is the *option* to green cheaply exactly when the crisis makes it optimal, and the browner economy enters with that option more fully intact. Pre-greening raises the steady-state green share and even the crisis switching volume; what it does not raise is the crisis *welfare* value of the margin. This is the sharpest form of the paper's thesis. The instruments that freeze the adoption technology, the cap here, and by construction the frozen-technology models of Bayer et al. (2026), Langot et al. (2026) and Auclert et al. (2024), miss an absorber that works best when left loaded, i.e. when the economy enters the crisis with room to green rather than already green.

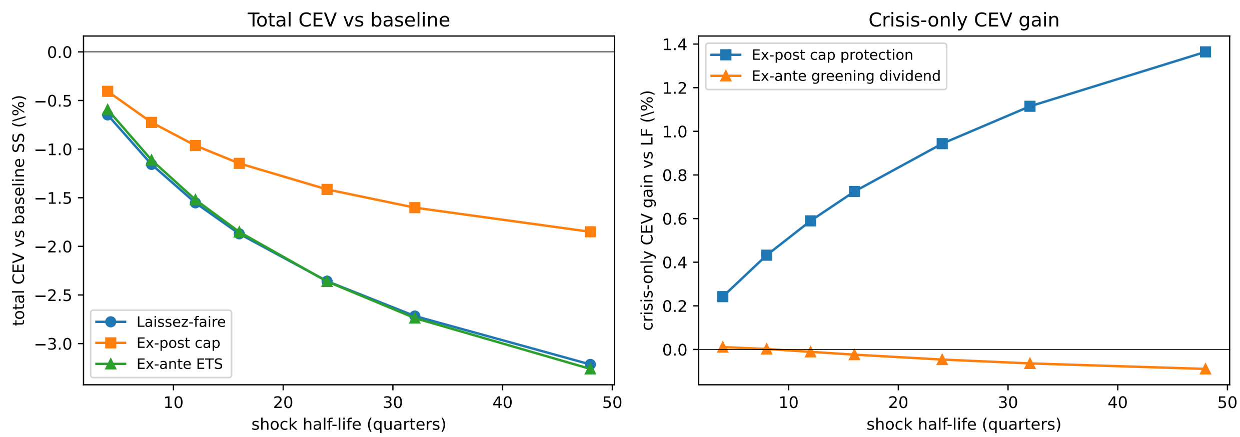

Figure 20: Persistence and the two policies. Left: total CEV of laissez-faire, the ex-post cap and the ex-ante ETS as the shock half-life rises. Right: the crisis-only gain over laissez-faire, the cap's protection grows with persistence, while the ex-ante *greening dividend* does not grow, it is zero to first order, because a more persistent shock lets the browner economy exploit the adoption absorber the prepared economy has already spent.

half-life	CEV_{LF}	CEV_{cap}	CEV_{ETS}	green div.	cap prot.
4	-0.648	-0.406	-0.594	0.0110	0.242
8	-1.158	-0.725	-1.112	0.0022	0.432
12	-1.553	-0.963	-1.520	-0.0108	0.590
16	-1.871	-1.147	-1.852	-0.0238	0.724
24	-2.358	-1.414	-2.361	-0.0462	0.944
32	-2.717	-1.602	-2.737	-0.0639	1.115
48	-3.214	-1.850	-3.260	-0.0891	1.364

Table 10: Route A: welfare by shock persistence (import booking, $\tau_b = 0.1$). Total CEV vs the no-ETS baseline SS. The crisis-only greening dividend (CEV_{ETS} minus the standing +0.043% minus CEV_{LF}) does not grow with persistence, while the cap protection does.

Takeaway. Three results point the same way. Energy-indexed targeting is dominated by a flat transfer of equal budget (Section 6.6); a permanent carbon tax is mildly welfare-improving in the steady state through a non-Pigouvian adoption correction, but essentially worthless as crisis insurance (Sections 7.1–7.2); and its insurance value does not grow, it is zero to first order, as shocks become persistent, because greening ex ante exhausts the adoption absorber (Section 7.4). The margin this paper adds is not a lever a planner should pre-pull; it is a stabiliser that pays off precisely when the economy is left free to adopt in the crisis.

8 Implications for European policy

The results map directly into the European policy architecture, on both the ex-post and the ex-ante side.

Shield incomes, not prices. The €600-billion-plus shield of 2021–23 was dominated by price-suppressing measures, caps, regulated tariffs, energy tax cuts (Bruegel, 2023). In the model, this is the instrument that buys output stabilisation at the cost of the entire crisis-time adoption wave: holding the brown retail price fixed removes the very signal that makes switching pay, and the extensive margin shuts (Section 6). The data bear the mechanism out. European heat pump sales, having surged 38% to a record three million units in 2022 when gas was expensive, fell 5% in 2023 and 21% in 2024 as gas prices normalised, and the industry's own post-mortem names "the low price of subsidised gas" among the three main causes (EHPA, 2024, 2025). A transfer of the same fiscal cost stabilises at least as well and leaves the wave intact. The 2022 episode should be read not only as a fiscal bill but as a forfeited adoption wave.

Pay the Social Climate Fund flat. The EU's carbon price for buildings and road transport (ETS2), scheduled to start in 2028 after a one-year postponement, comes attached to a Social Climate Fund that will mobilise at least €86.7 billion over 2026–2032, of which up to 37.5% may finance temporary direct income support to vulnerable households (European Commission, 2026). Our distributional result speaks to the design of that leg: because energy use tracks total consumption within technology groups, indexing support to energy expenditure targets wealth, not need. A flat payment of the same cost dominates the energy-indexed transfer in welfare, and the gain is concentrated exactly on the constrained households the Fund is meant to protect (Section 6). Income support that preserves the price signal is the model's best ex-post instrument; income support indexed to fossil-energy use is a weaker version of it.

Defend ETS2 as externality pricing, not insurance. A recurring argument for pre-crisis carbon pricing is resilience: a greener economy, it is said, is better insured against the next fossil shock. The model disciplines that claim. The crisis dividend of the prepared economy is zero to first order, shrinks with the size and persistence of the shock, and the mechanism is structural: pre-greening spends the adoption absorber, so the prepared economy enters the crisis with less welfare-relevant room to green, even though it switches more in volume (Section 7). The case for ETS2 is the externality and the standing welfare gain of the greener steady state, in the model, a permanent brown tax of roughly 10% is welfare-optimal on those grounds alone, not protection against the next crisis. Selling carbon pricing as insurance sets it up to disappoint.

9 Conclusion

We added a single margin, an endogenous brown/green durable adoption choice, to an otherwise standard small open economy HANK model, and asked how crisis policy interacts with it. The headline is a non-neutrality: a retail price cap stabilises output but, by holding the brown price fixed, switches the green adoption margin off, whereas a transfer of the same fiscal cost delivers comparable stabilisation and leaves the margin intact. Shielding consumers from the relative price is not neutral for the energy transition, and Europe's own heat-pump cycle (boom when gas was expensive, bust when subsidised gas made it cheap again) shows the margin operating in the data. The same logic, applied ex ante, yields the paper's sharpest result: a permanent carbon tax that greens the steady state provides no crisis insurance to first order and, as the shock becomes persistent, turns counterproductive, because pre-greening spends an adoption absorber that works best left loaded. The case for carbon pricing is the externality; the case for crisis protection is income support, paid flat.

The model is built to isolate one margin, and its limitations mark the road ahead. The taste scale σ_ε is calibrated to the quasi-experimental vehicle-adoption literature rather than estimated on European durables; the identification machinery of Section 4.1 makes the mapping transparent, and estimating the adoption elasticity directly is the obvious next step; heat-pump sales against the electricity–gas price ratio across European countries would be the natural design. The switching cost is booked as an import by accounting necessity under the inherited markup structure; the ordering of the policy results does not depend on this convention, but the sign of the adoption channel's output contribution does, and we have flagged it as conditional throughout (Appendix A). The green side of the economy is an accounting construct rather than a technology: there is no green supply sector with capacity and investment, and no installer income distribution beyond a lump-sum rebate. The supply side more broadly has no capital, which likely makes the transfer look somewhat more expansionary and the supply shock more inflationary than they would be with an investment margin. Finally, energy demand is homothetic conditional on the durable type, so the model is silent on the energy-share-of-the-poor mechanism of Bayer et al. (2026); the two distributional channels are complements, not substitutes.

None of these caveats touches what the paper establishes: the extensive margin of green adoption is a quantitatively live channel of energy-crisis policy, strong enough that the choice between shielding prices and shielding incomes is also a choice about the speed of the transition.

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Appendix

The appendices collect supporting material. Appendix A derives the balance-of-payments booking of the adoption flows and shows how the booking convention signs the adoption channel. Appendix B reports the accuracy of the linearised solution and the nonlinear welfare check. Appendix C develops the monetary analysis: the adoption margin leaves monetary transmission unchanged, while the monetary stance moves the adoption wave only weakly relative to the fiscal instruments. Appendix D treats the logit taste scale as both a numerical smoothing device and a structural adoption elasticity, and reports its identification.

A Balance-of-payments booking of the adoption flows

The sign and magnitude of the adoption channel's contribution to output depend on how the green-margin resource flows, the switching cost $\psi_g D^{sw}$ and the green energy bill, are booked in the balance of payments. The main text reports the import booking throughout. This appendix collects the two results that hinge on that choice: the domestic-booking sign comparison, and the carbon-revenue recycling experiment, which is naturally a booking-conditional statement about where the switching subsidy lands.

A.1 Import versus domestic booking of the adoption flows

Both statics of Section 5.2 are conditional on the import booking of the adoption flows. We solve the model under the alternative domestic booking of Section 3.4, green energy and green installation as competitive, zero-profit domestic value added rebated to households, with only brown energy imported, which resolves the phantom-dividend problem and clears to machine precision. Under this booking neither the switching cost nor the green energy bill is a leakage, and the adoption channel's contribution to cumulative output (Δ_{24}) flips sign, as Table 11 reports:

Δ_{24}, adoption output contribution	import booking	domestic booking
price shock	-8.79	$+28.56$
supply shock	-0.35	$+4.67$

Table 11: Adoption channel contribution to cumulative output (Δ_{24}), import vs. domestic booking (Section 3.4), against the common-steady-state counterfactual of Section 3. *None* policy.

A.2 Recycling the carbon revenue: rebate versus green subsidy

The welfare-optimal carbon tax of Section 7.1 returns its revenue lump sum. An alternative earmarks it to the transition itself, paying a share s_g of the green switching cost. We compare the two recyclings across carbon-tax rates $\tau_b \in \{0.10, 0.20, 0.30\}$, holding ψ_g at its no-tax value so the steady-state green share floats (Figures 21 and 22). Under the lump-sum rebate the steady-state green share rises from 8.9% to 20.1% across the grid, and the whole brown-price path in the crisis shifts up by the factor $(1 + \tau_b)$: the pre-tax price passes the world shock through identically, while households pay the tax-inclusive price. Under *budget-neutral* green-subsidy recycling, s_g solved so that the switching subsidy exactly exhausts the carbon revenue ($T^{\text{reb}} = 0$),the revenue funds a subsidy covering 35–40% of the switching cost and lifts the steady-state green share to 27–46%.

Two features are worth flagging. First, the carbon base is self-eroding: at $\tau_b = 0.10$ the revenue is a third smaller under green-subsidy recycling than under the rebate, because the greener economy holds less brown-durable stock to tax. Second, the crisis-time greening response is *smaller*, not larger, from the greener starting point under the subsidy: recycling that greens the economy through a lower switching *cost* preferentially converts the households nearest the logit margin, leaving a residual brown pool that is less responsive to the shock: the mirror image of the rebate, which greens through a higher permanent brown *price* and leaves the residual pool closer to the margin. This is the adoption-absorber exhaustion of Section 7.4 seen across policy instruments rather than across shock persistence.

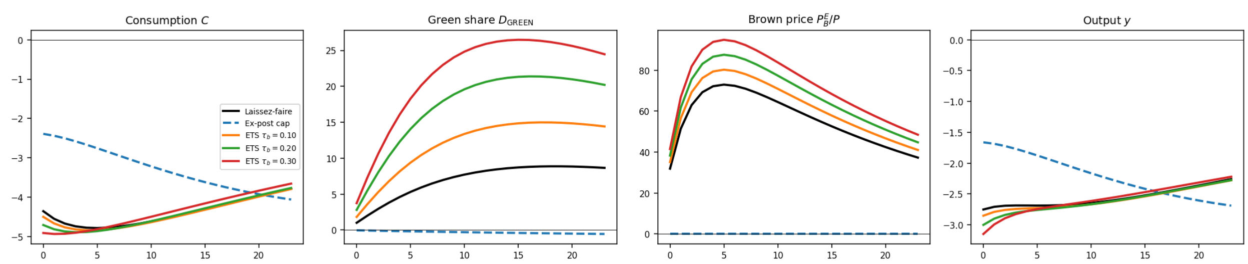

Figure 21: Carbon-tax sweep, lump-sum rebate: laissez-faire and ex-post cap against the ex-ante ETS economy at $\tau_b \in \{0.10, 0.20, 0.30\}$.

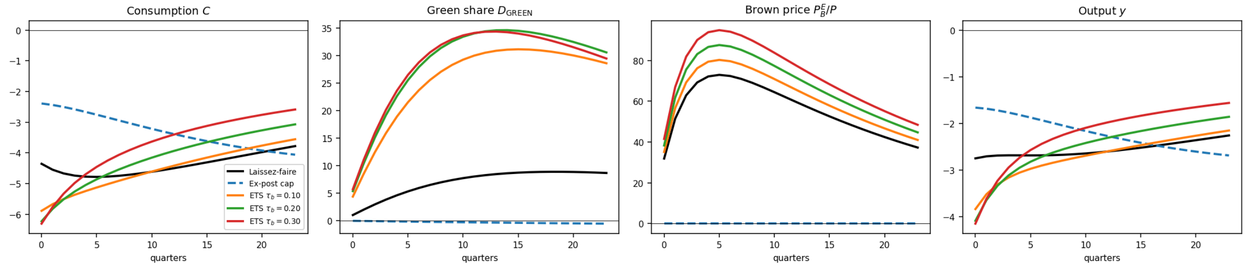

Figure 22: Carbon-tax sweep, budget-neutral green-subsidy recycling ($T^{\text{reb}} = 0$, s_g endogenous). Same axes as Figure 21.

B The linearised solution and the nonlinear welfare check

Two robustness exhibits that support the main-text results without being part of the argument: the accuracy of the linearised (sequence-space) solution given the discrete adoption choice, and the first-order welfare check on the nonlinear transition.

B.1 Robustness: the nonlinearity lives in the adoption margin

A natural concern is that the discrete adoption choice makes the economy too nonlinear for the linearised (sequence-space) solution to be reliable, in contrast to Auclert et al. (2024), who find nonlinearity immaterial in the continuous economy. Comparing the linearised impulse responses with their nonlinear counterparts (a fixed–Jacobian Newton solve) confines the concern to one variable. Aggregate output is essentially linear: the ratio of the nonlinear to the linear peak response, $y^{\text{NL}}/y^{\text{L}}$, stays in $[1.00, 1.01]$ across shock sizes, reproducing the small nonlinearity AMRS report. The nonlinearity is concentrated in the adoption share D^G, whose peak ratio rises monotonically from 1.16 at a one-eighth shock to 2.04 at a three-quarter shock (Figure 23, left). Freezing the margin does not reveal a more nonlinear economy underneath: under the common–steady-state counterfactual (Section 5.1), with D^G held fixed along the transition, output is itself only mildly nonlinear ($y^{\text{NL}}/y^{\text{L}} \approx 1.19$ at the largest shock), and the wedge between the nonlinear and linear paths is *wider* than with the margin open, not narrower. The discrete choice is where curvature concentrates, but leaving it free keeps aggregate output within one percent of its linear approximation.[2]

The convexity is the intrinsic curvature of a logit share evaluated near a low level. For a binary choice $P = [1 + \exp(-\Delta/\sigma_\varepsilon)]^{-1}$ in the value gap Δ, the relative curvature is

[2]At the baseline shock size ($\varepsilon = 1.0$) the nonlinear solution does not converge, so the sweep is reported to $\varepsilon = 0.75$. The D^G ratio is monotone and convex over $[0, 0.75]$, so nothing turns on the largest point.

$(1-2P)/\sigma_\varepsilon$, so $D^{G,\mathrm{NL}}/D^{G,\mathrm{L}} - 1 \propto 1/\sigma_\varepsilon$. Re-solving on a grid of σ_ε (holding $\bar{D}^G = 0.05$ by recalibrating ψ_g) confirms this: the ratio falls monotonically from 2.21 at $\sigma_\varepsilon = 0.02$ to 1.06 at $\sigma_\varepsilon = 0.40$ (Figure 23, right), and it tends to one as the shock vanishes, so the linearisation is exact to first order. Two implications follow. Aggregate and consumption-based welfare statistics are reliable under linearisation. The *level* of induced greening at the largest shocks is understated by the linear solution, and does not converge under a full-size shock, so adoption-channel magnitudes are read at a reduced shock size, and the taste scale σ_ε that governs this curvature is the same parameter whose identification Section 4.1 and Appendix D characterise: an adoption-elasticity moment pins both at once.

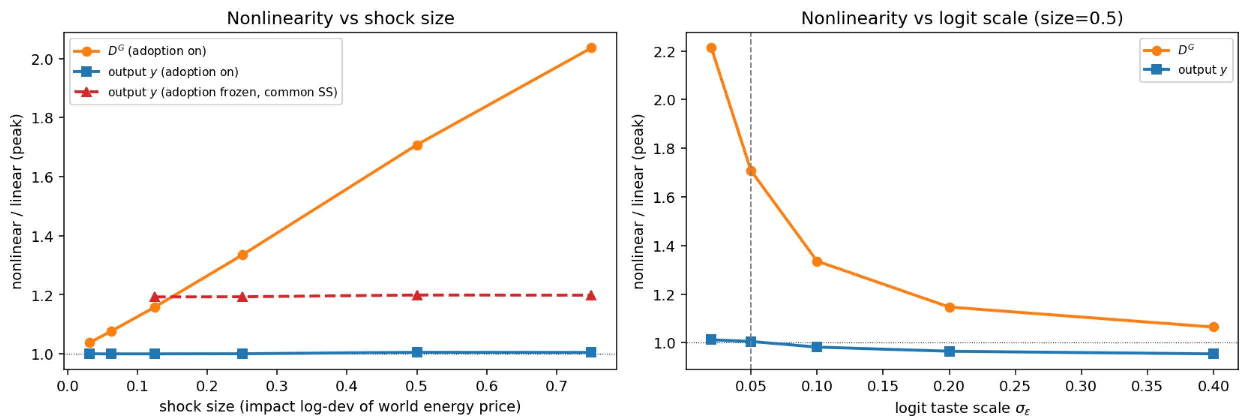

Figure 23: The nonlinearity is confined to the discrete adoption margin. *Left:* peak nonlinear/linear ratio vs shock size, output (adoption on and off) and D^G. *Right:* the same ratio vs the logit taste scale σ_ε at a half-size shock, ψ_g recalibrated to $\bar{D}^G = 5\%$.

B.2 First-order welfare and the nonlinear check

The consumption-equivalent variations reported throughout are first-order in the shock: flow utility along the linear impulse response is a first-order deviation of aggregate felicity, so the CEV is the leading term of the welfare expansion around a distorted steady state along a deterministic path. This is not the stochastic case in which first-order welfare evaluation is known to give spurious rankings (Kim and Kim, 2003; Benigno and Woodford, 2005): the first-order term is nonzero and dominant here. But the shock is large and the adoption margin is convex, so the size of the neglected second-order terms is an empirical question, which we settle by recomputing each scenario's welfare on the nonlinear perfect-foresight transition (Table 12).

size	Scenario	CEV% (linear)	CEV% (nonlinear)	NL/L	impat.	middle	patient
0.25	Laissez-faire	-0.4713	-0.4626	0.981	0.977	0.982	0.990
0.25	Ex-post cap	-0.2880	-0.2850	0.990	0.986	0.989	0.994
0.25	Slutsky transfer	-0.2047	-0.1984	0.969	0.947	0.967	0.989
0.25	Flat transfer	-0.0794	-0.0763	0.961	1.156	0.943	0.992
0.25	Ex-ante ETS	-0.4342	-0.4284	0.987	0.985	0.987	0.987
0.5	Laissez-faire	-0.9403	-0.8997	0.957	0.951	0.957	0.968
0.5	Ex-post cap	-0.5751	-0.5646	0.982	0.974	0.981	0.990
0.5	Slutsky transfer	-0.4089	-0.3769	0.922	0.877	0.916	0.962
0.5	Flat transfer	-0.1586	-0.1386	0.874	1.457	0.827	0.966
0.5	Ex-ante ETS	-0.9091	-0.8751	0.963	0.966	0.965	0.953
0.75	Laissez-faire	-1.4068	-1.3008	0.925	0.920	0.926	0.931
0.75	Ex-post cap	-0.8614	n.c.	–	–	–	–
0.75	Slutsky transfer	-0.6128	-0.5241	0.855	0.789	0.849	0.913
0.75	Flat transfer	-0.2377	-0.1749	0.736	1.892	0.654	0.915
0.75	Ex-ante ETS	-1.3816	-1.2825	0.928	0.939	0.934	0.899

Table 12: Welfare on the linear vs the nonlinear perfect-foresight transition, brown-price shock at reduced sizes (import booking). CEV is the total consumption-equivalent variation relative to the no-tax baseline steady state; NL/L is the ratio of nonlinear to linear CEV, overall and by discount-factor type. The headline shock (size 1.0) does not converge nonlinearly.

Three things follow. First, the linear CEV *overstates* the welfare loss, by 1–2% at a quarter-size shock and up to 7–26% at three-quarters, in the direction the convexity of the adoption margin predicts; the paper's levels are upper bounds on the losses, and would be overstated by an extrapolated 10–15% at the headline size, which does not converge nonlinearly. Second, the policy ordering, flat > Slutsky > cap > ETS > laissez-faire, holds at every converged size, linear and nonlinear alike, with the first-order gaps essentially unchanged; the welfare rankings stand. Third, the one difference that runs the other way is the impatient type under the flat transfer, whose nonlinear/linear ratio rises to 1.89: the linear solution understates the gain of high-MPC households when the transfer relaxes a binding liquidity constraint, which reinforces rather than weakens the distributional result of Section 6.6. The single claim the first-order calculation cannot support is the sign of the crisis-only greening dividend of Section 7.4, which is why we report it there as zero to first order.

C Monetary policy and the adoption margin

Two questions connect monetary policy to the adoption margin, and they point in opposite directions. Does the margin reshape how a monetary shock transmits? And does the monetary stance, during an energy shock, reshape the adoption wave? The first is a robustness check; the second is where the interest-rate sensitivity of green durables that Dietrich et al. (2025) identify in a representative agent shows up in our setting.

C.1 The adoption margin does not reshape monetary transmission

The energy-shock experiments hold monetary policy at the AMRS constant-real-rate baseline. A 100-basis-point (annualised) tightening isolates the role of the rule itself. Figure 24 contrasts the constant-real-rate baseline with a standard Taylor rule $(\phi_\pi, \phi_{\pi e}) = (1.5, 0)$. The response is the textbook one: output and consumption fall, inflation falls. It is *not* materially reshaped by the adoption margin. At a 5% green steady state the durable-choice channel is too small to move aggregate monetary transmission (on impact, output -0.86 under the Taylor rule against -1.13 under the constant-real-rate rule; inflation -0.42 against -0.48 annualised). The adoption margin matters for how the economy absorbs an *energy* shock, not for how it transmits a *monetary* one.

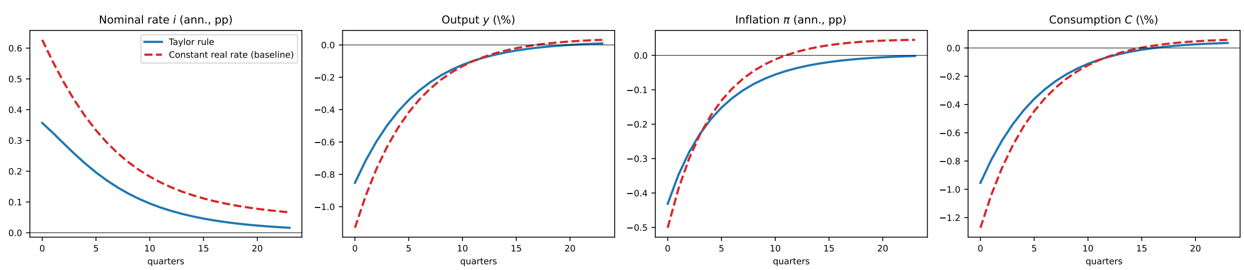

Figure 24: Monetary-policy shock (100bp annualised tightening): Taylor rule (solid) versus the constant-real-rate baseline (dashed). Nominal rate, output, inflation, consumption.

C.2 The monetary stance moves the wave, but weakly

The reverse question is Dietrich's. In their representative-agent economy green durables are interest-rate sensitive, so a strict inflation target slows the transition. The mechanism survives here and acquires a distributional edge: the switch is a lumpy purchase financed against the borrowing constraint, so its user cost rises with the real rate precisely for the constrained households that account for most of the crisis adoption (Figure 1).

The stance operates on the wave, but weakly. Table 13 holds the brown-price shock fixed and varies the rule. A Taylor rule that leans against the energy-driven inflation ($\phi_\pi = 1.5$)

raises the ex-ante real rate, by +4.5 cumulated over $H = 24$, and shrinks the peak green-share response from 7.72 to 7.65 pp. A deliberate 25-basis-point accommodation does the opposite, lifting it to 7.8 pp and cutting the output loss; the standalone rate cut raises adoption on its own (D^G peak +0.11 pp, consumption +5.8 cumulated) as easier financing relaxes the constraint on the switch. The sign is Dietrich's; the magnitude is small.

stance	peak ΔD^G	$\sum D^{sw}$	$\sum y$	$\sum C$	$\sum r$
Constant real rate (baseline)	7.72	14.43	-64.7	-108.9	+0.00
Active Taylor $\phi_\pi = 1.5$	7.65	14.28	-76.0	-125.1	+4.52
Baseline + 25bp accommodation	7.79	14.57	-59.0	-103.1	-1.55

Table 13: Monetary stance and the adoption wave under the brown-price shock. Peak green-share response and cumulative switching, output, consumption and the ex-ante real rate ($\times 100$, $H = 24$). A Taylor rule that leans against the energy-driven inflation raises the real rate and modestly shrinks the wave at a large output cost; a deliberate accommodative innovation amplifies it.

Figure 25 sweeps the inflation coefficient. Stricter targeting shrinks the wave monotonically, from 7.74 pp at $\phi_\pi = 1.25$ to 7.44 at $\phi_\pi = 3$, but the whole range spans a third of a percentage point, while the cumulative output loss grows from -70 to -92. Strict inflation targeting during an energy crisis buys a marginal slowdown of the transition at a first-order output cost.

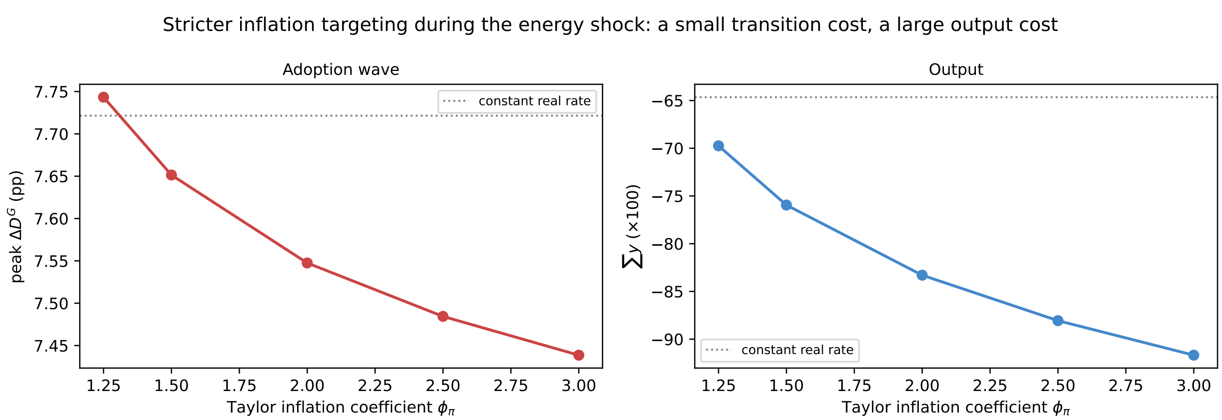

Figure 25: The monetary stance and the adoption wave under the brown-price shock, as the Taylor inflation coefficient ϕ_π rises (with $\phi_{\pi e} = 0$). Left: peak ΔD^G. Right: cumulative output. The dotted line is the constant-real-rate baseline. Stricter targeting shrinks the wave marginally and deepens the output loss substantially.

The comparison that matters is with the fiscal instruments. The peak wave's semi-elasticity to the real rate is about -0.015 pp per unit of cumulative rate, so moving the wave by one percentage point would take a cumulative rate change of order sixty, an implausible monetary action, whereas the retail price cap removes almost the entire 7.7 pp wave at a stroke (Section 6.1). Monetary policy reaches the adoption margin only through the financing cost of the switch, a second-order channel; the cap and the transfer reach it through the relative price itself, which is first-order. This is why we keep monetary policy at the constant-real-rate baseline in the main text: not because the stance is neutral for adoption, but because it is dominated, by an order of magnitude, by the instruments the paper ranks.

An accommodative offset. A natural crisis exercise runs an accommodative innovation *alongside* the energy shock, leaning against it rather than waiting. Figure 26 sizes the accommodation to halve the cumulative output loss, which takes a large front-loaded cut (about 144 basis points annualised). It cushions cumulative output (-65 to -32) and consumption (-109 to -76) and lifts the adoption wave (7.7 to 8.1 pp), the financing channel working in its favour, but pays for the relief with more inflation (impact annualised inflation +5.6% to +8.3%) and an output path that overshoots on impact before settling. The offset buys output stabilisation and a small transition dividend at the cost of the price level and a much larger rate move than the fiscal instruments require; it does not substitute for the relative-price instruments, which stabilise output without that inflation cost.

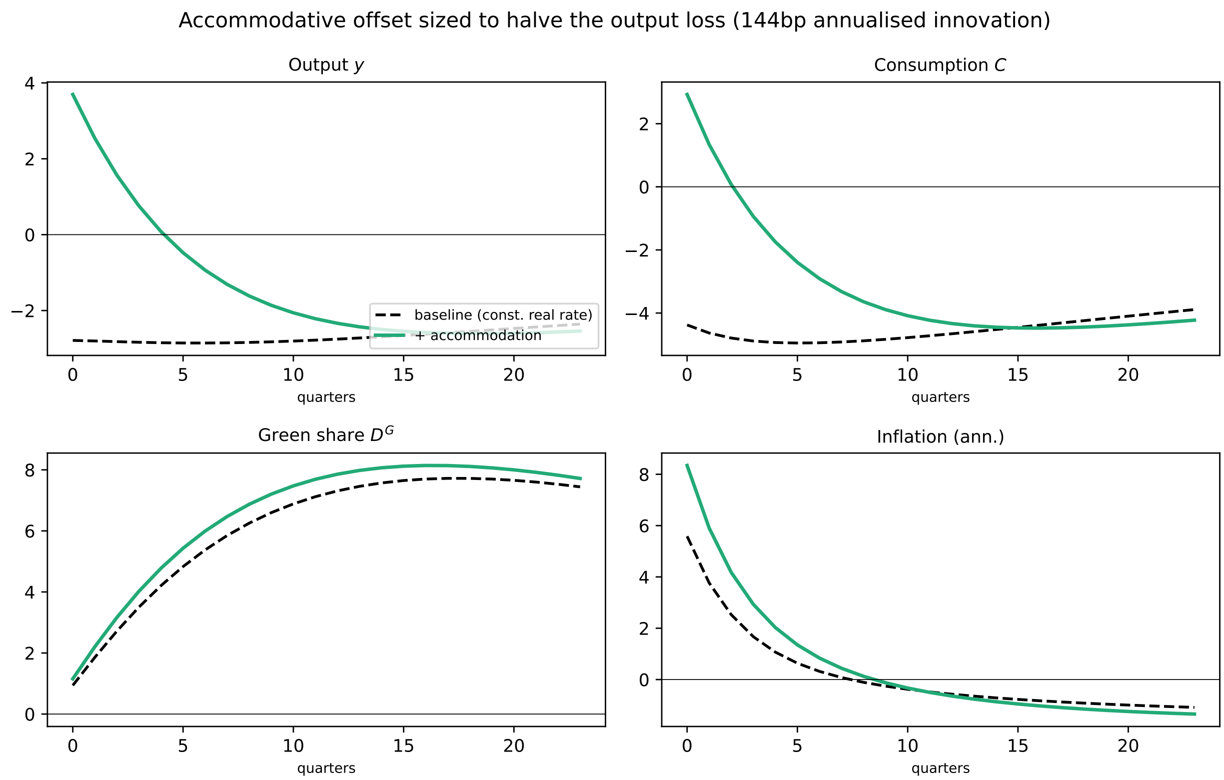

Figure 26: Accommodative monetary offset run alongside the brown-price shock, sized to halve the cumulative output loss: energy shock alone (dashed) versus energy shock plus accommodation (solid). The offset cushions output and consumption and amplifies adoption, at the cost of higher inflation. Level deviations $\times 100$.

D The logit taste scale: smoothing device and structural elasticity

The extreme-value taste shocks on the adoption choice play two roles at once, and this appendix separates them. The first role is computational: the logit smooths the discrete policy, so that choice probabilities and the value function are differentiable with closed-form derivatives, which is what the sequence-space method requires to build Jacobians through the household block (Iskhakov et al., 2017). In that tradition the scale σ_ε is a numerical device: one takes it small and verifies that results are insensitive to it. The second role is economic. Differentiating the logit gives $dP_i = P_i\,(dV_i - dV)/\sigma_\varepsilon$: the response of the choice probabilities to *any* perturbation of the value gap scales with $1/\sigma_\varepsilon$, so the same parameter that smooths the policy is the structural elasticity of the margin. Which of the two readings governs is not a matter of taste; it depends on where the population sits on the logit.

If choice probabilities are interior, of order one, as for labor-force participation, where employment rates and quarterly flows are tens of percent, then $P(1-P)$ is bounded, the $\sigma_\varepsilon \to 0$ limit is a well-defined deterministic cutoff, and aggregate responses converge to the density of households at the threshold: an object of the wealth and productivity distribution, not of σ_ε. There the numerical reading is legitimate, the economics of the margin is carried by the distribution (and, in search applications, by the frictions standing between the choice and the outcome), and the smoothing doctrine applies. If instead the margin is *rare*, choice probabilities deep in the left tail, the logit is exponential there, $d \log P = d\Delta/\sigma_\varepsilon$ holds for every household regardless of its level, and the semi-elasticity of the flow is identically $1/\sigma_\varepsilon$. No feature of the distribution can attenuate it, there is no σ_ε-insensitive range, and the scale *is* the elasticity of the margin.

Our margin is rare, verifiably so. The calibration targets a 5% green share with a five-year durable life, which fixes the steady-state switching flow at $\delta_g \bar{D}^G/(1-\bar{D}^G) = 0.26\%$ per quarter. Table 14 reports the distribution-weighted quantiles of the quarterly switch probability among brown incumbents: the mean is 0.26%, the median is essentially zero, the 95th percentile is 1.6%, and even the 99.9th percentile stays below 9%. Virtually no mass approaches interior probabilities. This is the world of rare, lumpy, quasi-irreversible adjustment, the regime in which the structural discrete-choice literature estimates the scale from choice data (Rust, 1987), not the world of participation.

type	β	mass%	mean	p50	p95	p99.9	max
impatient	0.924	95.9	0.23	0.00	1.58	8.74	26.4
middle	0.954	95.2	0.26	0.00	1.66	8.62	23.4
patient	0.984	94.6	0.30	0.03	1.63	6.82	17.4
all	–	95.2	0.26	0.00	1.58	8.45	26.4

Table 14: Quarterly switch probabilities among brown incumbents (distribution-weighted, baseline steady state). Mean, median and upper quantiles of $P(\text{switch})$ in %, by discount-factor type and pooled. The mean equals the steady-state flow identity $\delta_g \bar{D}^G/(1-\bar{D}^G) = 0.26\%$; virtually no mass approaches interior probabilities, so the brown population sits in the exponential tail of the logit (Appendix D).

Three consequences follow, each handled in the paper. First, the adoption-channel magnitudes are proportional to $1/\sigma_\varepsilon$, so a level moment cannot discipline them; Section 4.1 characterises the $(\sigma_\varepsilon, \psi_g)$ locus and identifies the adoption elasticity as the moment that pins the scale. Second, the same $1/\sigma_\varepsilon$ governs the curvature of the margin, so the linearisation error is concentrated in D^G and grows as σ_ε falls; Appendix B.1 quantifies it. Third, the steady state and the policy ordering are invariant along the locus, so none of the paper's qualitative results turn on the scale. What does turn on it, the size of the adoption wave and of the adoption channel, is exactly what the elasticity moment disciplines.

D.1 From micro estimates to the taste scale: the procedure

This subsection documents the calibration of σ_ε step by step, so that every number in Section 4.1 can be audited and reproduced. The entire procedure is implemented in a single runner (`runners/run_taste_identification.py`; `python main.py taste_id` regenerates Table 15, Table 14 and Figure 27); the grid, the perturbation sizes and the dollar mapping are constants at the top of the file.

1. *Trace the locus.* For each σ_ε on the grid $\{0.02, 0.035, 0.05, 0.07, 0.10, 0.15, 0.20\}$, resolve the baseline steady state with ψ_g as the unknown targeting $\bar{D}^G = 5\%$. This delivers $\psi_g^*(\sigma_\varepsilon)$, rising from 0.32 to 1.53, with the steady state, shares, flows, prices, distribution, invariant along the locus.

2. *Switch to the fixed-ψ configuration.* All comparative statics below hold ψ_g fixed at ψ_g^* and let the green share D^G adjust endogenously (the same configuration that builds the ex-ante ETS economy of Section 7.2). Elasticities are computed as finite differences on this configuration around the calibrated point.

3. *Operating-cost margin.* Perturb the delivered green/brown price ratio $\varrho = 0.80$ down by 0.01 at fixed ψ_g^* and measure $\Delta \ln D^G / \Delta \ln(1-\varrho)$, the elasticity of the green share to the operating-cost gap. It falls from 2.8 to 0.4 across the grid (1.32 at the baseline).

4. *Upfront-cost margin: the matched moment.* Perturb ψ_g down by 1% at fixed calibration and measure the response of the switching flow $P = D^{sw}/(1-D^G)$. Per 1% of ψ_g the semi-elasticity is nearly flat in σ_ε (5.1% to 3.8%), because ψ_g^* rises along the locus and compensates; the identifying variation comes from converting to absolute dollars. With one model unit equal to quarterly household consumption $\bar{C}_\$$, the semi-elasticity per \$1,000 is the per-1% figure times $1{,}000/(0.01\,\psi_g^*\,\bar{C}_\$)$; at $\bar{C}_\$ = \$20{,}000$ it falls from 80% to 12% across the grid (Table 15).

The empirical anchors are converted to the same units. Gallagher and Muehlegger (2011) and Chandra et al. (2010) estimate that a \$1,000 tax incentive raises hybrid adoption by 31–38%; no conversion is needed. Muehlegger and Rapson (2022) report a demand elasticity of -2.1 at a mean transaction price of \$26,000, which converts to $2.1 \times 1{,}000/26{,}000 \approx 8\%$ per \$1,000; their estimated pass-through of 73–85% implies the price faced by the buyer moves nearly one-for-one with the subsidy, so the conversion needs no incidence correction.

The choice between the two anchors is a population statement, not a taste: at a 5% green share the model's marginal adopters consume roughly twice the average brown household, they are the higher-income early adopters of the hybrid-era estimates, not the mass-market population of Muehlegger and Rapson (2022), whose estimate therefore serves as the floor on responsiveness rather than the target.

Matching the early-adopter range pins $\sigma_\varepsilon \in [0.05, 0.07]$. The implied premium $\psi_g^* \bar{C}_\$$ provides an independent cross-check: it is \$11,000–\$14,000 on the pinned interval, within the heat-pump/EV premium range, while $\sigma_\varepsilon \geq 0.10$ would imply premia of \$18,000 and above.

The one free constant is the dollar mapping $\bar{C}_\$$, an assumption stated, not estimated. Its role is transparent: a higher $\bar{C}_\$$ scales down the model's per-\$1,000 response and shifts the elasticity-based pin toward lower σ_ε, while scaling up the implied premium and shifting the premium-based cap in the same direction, so the two criteria co-move and the joint interval stays tight. Concretely, at $\bar{C}_\$ = \$16{,}000$ the joint pin is roughly $[0.07, 0.10]$; at \$20,000, $[0.05, 0.07]$; at \$24,000, $[0.04, 0.06]$. Every one of these intervals lies inside the sensitivity grid of Table 15, over which the steady state and the policy ordering are invariant; what the mapping moves is the size of the adoption channel, bounded on the grid by peak responses of 2 to 20 pp.

Two limits of the procedure are worth stating. The anchors come from vehicle adoption in the United States in the 2000s–2010s; their external validity for heat pumps or for European households is an assumption, and estimating the adoption elasticity directly, rather than importing it, is the natural next step (Section 9). And the anchors identify the upfront-cost margin; the operating-cost margin, the channel documented by Beresteanu and Li (2011) for gasoline prices and hybrid demand, is disciplined here only through the model's internal consistency, which ties the two margins together at the calibrated point.

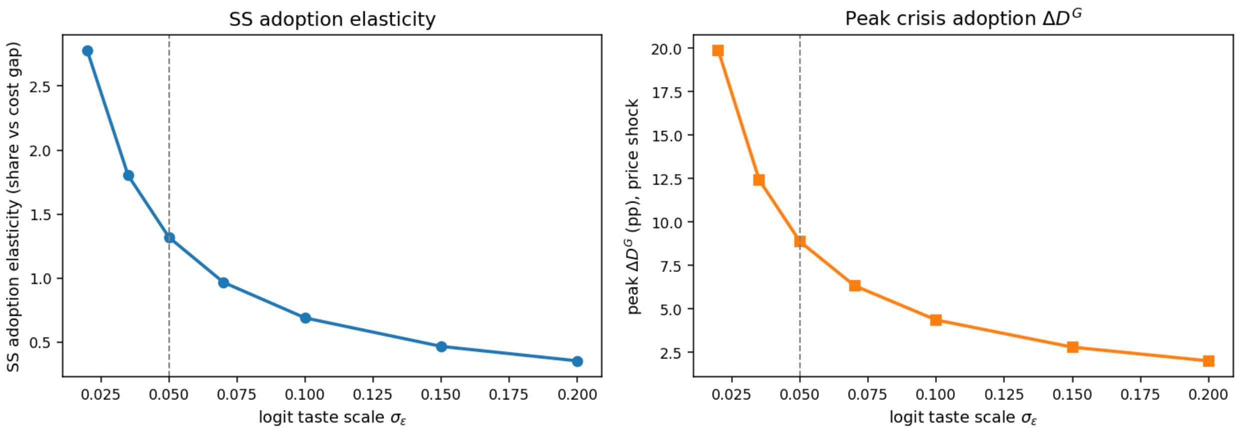

Figure 27: Identification of the taste scale. *Left*: the steady-state adoption elasticity (green share vs. the operating-cost gap) against σ_ε. *Right*: the crisis adoption magnitude (peak ΔD^G to the brown-price shock). The steady state is held at $\bar{D}^G = 5\%$ by recalibrating ψ_g; the dashed line marks the baseline $\sigma_\varepsilon = 0.05$.

σ_ε	ψ_g	premium \$000	cost-gap elast.	%/\$1,000	peak ΔD^G	$\sum y$
0.020	0.323	6.5	2.78	80	19.86	−59.2
0.035	0.430	8.6	1.80	54	12.42	−60.9
0.050	0.538	10.8	1.32	40	8.87	−61.5
0.070	0.680	13.6	0.97	31	6.34	−61.6
0.100	0.888	17.8	0.69	22	4.36	−61.4
0.150	1.220	24.4	0.47	16	2.80	−60.6
0.200	1.534	30.7	0.36	12	2.01	−59.8

Table 15: Identification of the logit taste scale. For each σ_ε, ψ_g is recalibrated to hold $\bar{D}^G = 5\%$, leaving the steady state invariant. Columns: the implied green premium in dollars (one unit = \$20,000 quarterly consumption); the steady-state elasticity of the green share to the operating-cost gap at fixed ψ_g; the semi-elasticity of the switching flow to a \$1,000 cut in ψ_g, matched to Gallagher and Muehlegger (2011); Chandra et al. (2010); Muehlegger and Rapson (2022); the peak crisis adoption response; and the cumulative output loss, flat while the adoption responsiveness varies by an order of magnitude.

E Calibrating the operating-cost ratio ϱ

The single price parameter of the adoption block, $\varrho = \bar{P}^E_G / \bar{P}^E_B$, is a delivered operating cost: the cost of one unit of energy service (a kilometre) from the green durable relative to the brown one. It is not a price per unit of energy (per kilowatt-hour, electricity is dearer than gasoline) but a cost per unit of service, net of each technology's conversion efficiency. The upfront cost of switching is a separate object, ψ_g.

We calibrate ϱ on the electric-vehicle margin, the durable for which both the operating-cost gap and the adoption elasticity (Section 4.1) are measured. With P^G, P^B the retail electricity and gasoline prices and e_G, e_B real-world energy use,

$$\varrho = \frac{P^G\, e_G}{P^B\, e_B} = \frac{0.290 \times 0.21}{1.62 \times 0.070} = \frac{0.0609}{0.1134} = 0.54.$$

		Value	Source
Electricity price, households	P^G	0.290 EUR/kWh	Eurostat `nrg_pc_204`, 2025S2, incl. taxes
Gasoline price, Euro-95	P^B	1.62 EUR/L	EC Weekly Oil Bulletin, 2025S2, incl. taxes
Electric-car energy use	e_G	0.21 kWh/km	IEA *Global EV Outlook 2026*
Gasoline-car fuel use	e_B	7.0 L/100km	National on-road data (e.g. FR SDES)

Table 16: Inputs to ϱ. European, tax-inclusive, same semester.

Three choices are worth stating. *(i) Tax-inclusive prices.* The baseline has no carbon tax, so the household compares retail prices; the gap between the two ratios (0.54 inclusive against 0.88 excluding taxes) is the asymmetric taxation of the two fuels, which the ETS acts on rather than assumes away. *(ii) Real-world energy use.* Both figures are on-road or metered rather than type-approval, so charging losses are already in the electricity figure and the two sides rest on the same basis. *(iii) Fleet averages.* A matched compact pair (electric 17 kWh/100km, gasoline 6.2 L/100km) gives 0.49, so the ratio does not turn on vehicle-size composition. As an internal check, 0.21 kWh/km is 2.3 litres of gasoline equivalent per 100 km, 68% below the gasoline car's 7.0, matching the roughly 70% energy saving the same source reports for a comparable electric car. The heating margin (an air-source heat pump against a condensing gas boiler at Eurostat household electricity and gas prices) gives a nearby $\varrho \approx 0.62$, so reading the durable as a heat-pump/EV composite would move the ratio only inside the range below.

Sensitivity. Holding the steady-state green share at $\bar{D}^G = 5\%$, ψ_g re-solves as ϱ varies. Across the whole plausible range the policy CEV ranking is unchanged (flat transfer $\succ$ Slutsky $\succ$ ex-post cap $\succ$ ex-ante ETS $\succ$ laissez-faire $\succ$ green subsidy); only the crisis adoption magnitude responds, by about one percentage point of peak green share.

ϱ	0.48	0.54	0.62	0.70	0.80
ψ_g (solved)	0.827	0.770	0.696	0.625	0.538
peak ΔD^G (pp)	7.44	7.72	8.10	8.46	8.87
peak output loss $-y$ (%)	2.85	2.86	2.87	2.88	2.90

Table 17: Sensitivity of the baseline to ϱ. Price shock, import booking. $\bar{D}^G = 5\%$ imposed throughout; the policy ordering (not shown) is invariant.